

NONLINEAR WAVE THEORY SEMINAR NOTES

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ABSTRACT. These notes collect selected computations and seminar writeups for the course *Nonlinear Wave Theory*, delivered by Professor Liming Ling in Spring 2026.

The full L^AT_EX source code of these notes is available at: <https://codeberg.org/killllakill/courseNotes>.

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HW 1: Two-Step Gauge Transformation

2026-04-22

Consider the system

$$(1.1) \quad i\Psi_t = \frac{1}{2}Q^2\Psi - (\Psi^\dagger\Psi)\Psi + \frac{\Delta}{2}\sigma_3\Psi, \quad \Psi = (\Psi_1, \Psi_2)^\top,$$

$$(1.2) \quad Q = -i\partial_x + \alpha A(x), \quad A(x) = \begin{pmatrix} 0 & e^{-2ikx} \\ e^{2ikx} & 0 \end{pmatrix}.$$

Here

$$(1.3) \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

denote the Pauli matrices.

We first compute the square of Q . Since

$$(1.4) \quad A(x)^2 = \mathbb{I}_2, \quad A_x(x) = \begin{pmatrix} 0 & -2ik e^{-2ikx} \\ 2ik e^{2ikx} & 0 \end{pmatrix},$$

we obtain

$$\begin{aligned} Q^2 &= (-i\partial_x + \alpha A)^2 \\ &= -\partial_x^2 - i\alpha A\partial_x - i\alpha\partial_x(A\cdot) + \alpha^2 A^2 \\ &= -\partial_x^2 - i\alpha A\partial_x - i\alpha(A_x + A\partial_x) + \alpha^2 A^2 \\ (1.5) \quad &= -\partial_x^2 - 2i\alpha A\partial_x - i\alpha A_x + \alpha^2. \end{aligned}$$

Substituting (1.5) into (1.1) yields

$$(1.6) \quad i\Psi_t = -\frac{1}{2}\Psi_{xx} - i\alpha A\Psi_x - \frac{i}{2}\alpha A_x\Psi + \frac{1}{2}\alpha^2\Psi - |\Psi|^2\Psi + \frac{\Delta}{2}\sigma_3\Psi.$$

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Define the gauge transformation

$$(1.7) \quad \Psi(x, t) = \Lambda(x, t)\psi(x, t), \quad \Lambda(x, t) = e^{-\frac{i}{2}(\alpha^2 + k^2)t} e^{-ikx\sigma_3}.$$

Since $\sigma_3^2 = \mathbb{I}_2$, the matrix exponential may be written as

$$(1.8) \quad e^{-ikx\sigma_3} = \begin{pmatrix} e^{-ikx} & 0 \\ 0 & e^{ikx} \end{pmatrix},$$

and therefore Λ is unitary:

$$(1.9) \quad \Lambda^\dagger \Lambda = \mathbb{I}_2.$$

Consequently,

$$(1.10) \quad |\Psi|^2 = \Psi^\dagger \Psi = \psi^\dagger \Lambda^\dagger \Lambda \psi = \psi^\dagger \psi = |\psi|^2.$$

Differentiating (1.7) gives

$$(1.11) \quad \Lambda_t = -\frac{i}{2}(\alpha^2 + k^2)\Lambda, \quad \Lambda_x = -ik\sigma_3\Lambda.$$

Hence

$$(1.12) \quad \Psi_t = \Lambda_t \psi + \Lambda \psi_t = \Lambda \left(-\frac{i}{2}(\alpha^2 + k^2)\psi + \psi_t \right),$$

$$(1.13) \quad \Psi_x = \Lambda_x \psi + \Lambda \psi_x = \Lambda (-ik\sigma_3 \psi + \psi_x),$$

and

$$(1.14) \quad \begin{aligned} \Psi_{xx} &= \partial_x [\Lambda (-ik\sigma_3 \psi + \psi_x)] \\ &= \Lambda_x (-ik\sigma_3 \psi + \psi_x) + \Lambda (-ik\sigma_3 \psi_x + \psi_{xx}) \\ &= \Lambda [-ik\sigma_3 (-ik\sigma_3 \psi + \psi_x) - ik\sigma_3 \psi_x + \psi_{xx}] \\ &= \Lambda (-k^2 \psi - 2ik\sigma_3 \psi_x + \psi_{xx}). \end{aligned}$$

Next we compute the conjugation of $A(x)$ by Λ . Since the scalar phase $e^{-\frac{i}{2}(\alpha^2 + k^2)t}$ commutes with every matrix, we have

$$(1.15) \quad \begin{aligned} \Lambda^{-1} A(x) \Lambda &= e^{ikx\sigma_3} A(x) e^{-ikx\sigma_3} \\ &= \begin{pmatrix} e^{ikx} & 0 \\ 0 & e^{-ikx} \end{pmatrix} \begin{pmatrix} 0 & e^{-2ikx} \\ e^{2ikx} & 0 \end{pmatrix} \begin{pmatrix} e^{-ikx} & 0 \\ 0 & e^{ikx} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_1. \end{aligned}$$

Similarly,

$$(1.16) \quad \Lambda^{-1} A_x(x) \Lambda = \begin{pmatrix} 0 & -2ik \\ 2ik & 0 \end{pmatrix} = 2k \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = 2k\sigma_2 = 2ik\sigma_1\sigma_3.$$

Moreover,

$$(1.17) \quad \Lambda^{-1} \sigma_3 \Lambda = \sigma_3,$$

since σ_3 commutes with $e^{-ikx\sigma_3}$.

Substituting (1.12), (1.13), and (1.14) into (1.6), we obtain

$$\begin{aligned}
& i\Lambda \left(-\frac{i}{2}(\alpha^2 + k^2)\psi + \psi_t \right) \\
&= -\frac{1}{2}\Lambda \left(-k^2\psi - 2ik\sigma_3\psi_x + \psi_{xx} \right) \\
&\quad - i\alpha A\Lambda (-ik\sigma_3\psi + \psi_x) - \frac{i}{2}\alpha A_x\Lambda\psi + \frac{1}{2}\alpha^2\Lambda\psi \\
&\quad - |\psi|^2\Lambda\psi + \frac{\Delta}{2}\sigma_3\Lambda\psi.
\end{aligned}
\tag{1.18}$$

Multiplying (1.18) from the left by Λ^{-1} and using (1.15)–(1.17) gives

$$\begin{aligned}
\frac{1}{2}(\alpha^2 + k^2)\psi + i\psi_t &= \frac{1}{2}k^2\psi + ik\sigma_3\psi_x - \frac{1}{2}\psi_{xx} \\
&\quad - i\alpha\sigma_1(-ik\sigma_3\psi + \psi_x) - i\alpha k\sigma_2\psi \\
&\quad + \frac{1}{2}\alpha^2\psi - |\psi|^2\psi + \frac{\Delta}{2}\sigma_3\psi.
\end{aligned}
\tag{1.19}$$

Using the identity

$$(1.20) \quad \sigma_1\sigma_3 = -i\sigma_2,$$

we find

$$(1.21) \quad -i\alpha\sigma_1(-ik\sigma_3\psi) = i\alpha k\sigma_2\psi,$$

which exactly cancels the term $-i\alpha k\sigma_2\psi$ in (1.19). After cancellation of the constant terms, we arrive at

$$(1.22) \quad i\psi_t = -\frac{1}{2}\psi_{xx} + i(k\sigma_3 - \alpha\sigma_1)\psi_x - |\psi|^2\psi + \frac{\Delta}{2}\sigma_3\psi.$$

We now introduce the second gauge transformation

$$(1.23) \quad \psi(x, t) = V(x, t)q(x, t), \quad V(x, t) = e^{\frac{i}{2}(\alpha^2 + k^2)t} B e^{-ikx\sigma_3}, \quad \kappa = \sqrt{\alpha^2 + k^2},$$

where

$$(1.24) \quad B = \begin{pmatrix} -\sin \varphi & -\cos \varphi \\ \cos \varphi & -\sin \varphi \end{pmatrix} = -\sin \varphi \mathbb{I}_2 + \cos \varphi \sigma_1\sigma_3.$$

The angle φ is chosen so that

$$(1.25) \quad 2\varphi = \arctan \frac{k}{\alpha} - \frac{\pi}{2}.$$

Hence

$$(1.26) \quad \tan(2\varphi) = -\frac{\alpha}{k},$$

and therefore

$$(1.27) \quad \sin(2\varphi) = -\frac{\alpha}{\sqrt{\alpha^2 + k^2}}, \quad \cos(2\varphi) = \frac{k}{\sqrt{\alpha^2 + k^2}}.$$

Since B is independent of x and t , differentiating (1.23) yields

$$(1.28) \quad \psi_t = V \left(\frac{i}{2}\kappa^2 q + q_t \right),$$

$$(1.29) \quad \psi_x = V (-ik\sigma_3 q + q_x),$$

$$(1.30) \quad \psi_{xx} = V \left(-\kappa^2 q - 2ik\sigma_3 q_x + q_{xx} \right).$$

Next we compute several algebraic identities associated with B . Since

$$(1.31) \quad B^\top = \begin{pmatrix} -\sin \varphi & \cos \varphi \\ -\cos \varphi & -\sin \varphi \end{pmatrix},$$

a direct computation gives

$$(1.32) \quad B^\top B = \mathbb{I}_2.$$

Consequently,

$$(1.33) \quad V^\dagger V = \mathbb{I}_2,$$

and hence the nonlinear term is preserved:

$$(1.34) \quad |\psi|^2 = \psi^\dagger \psi = q^\dagger V^\dagger V q = q^\dagger q = |q|^2.$$

We now compute the conjugation of σ_3 by B . Using (1.27), one finds

$$(1.35) \quad B^\top \sigma_3 B = \begin{pmatrix} -\cos(2\varphi) & \sin(2\varphi) \\ \sin(2\varphi) & \cos(2\varphi) \end{pmatrix} = -\frac{1}{\sqrt{\alpha^2 + k^2}} \begin{pmatrix} k & \alpha \\ \alpha & -k \end{pmatrix}.$$

Using (1.24), we compute

$$(1.36) \quad \begin{aligned} B^\top (k\sigma_3 - \alpha\sigma_1) B &= kB^\top \sigma_3 B - \alpha B^\top \sigma_1 B \\ &= \begin{pmatrix} -k\cos(2\varphi) + \alpha\sin(2\varphi) & k\sin(2\varphi) + \alpha\cos(2\varphi) \\ k\sin(2\varphi) + \alpha\cos(2\varphi) & k\cos(2\varphi) - \alpha\sin(2\varphi) \end{pmatrix}. \end{aligned}$$

Substituting (1.27) gives

$$(1.37) \quad B^\top (k\sigma_3 - \alpha\sigma_1) B = \begin{pmatrix} -\kappa & 0 \\ 0 & \kappa \end{pmatrix} = -\kappa\sigma_3.$$

Substituting (1.28)–(1.30) into (1.22), we obtain

$$(1.38) \quad \begin{aligned} iV \left(\frac{i}{2} \kappa^2 q + q_t \right) &= -\frac{1}{2} V \left(-\kappa^2 q - 2i\kappa\sigma_3 q_x + q_{xx} \right) \\ &\quad + i(k\sigma_3 - \alpha\sigma_1) V (-i\kappa\sigma_3 q + q_x) - |q|^2 V q + \frac{\Delta}{2} \sigma_3 V q. \end{aligned}$$

Multiplying (1.38) from the left by V^{-1} yields

$$(1.39) \quad \begin{aligned} -\frac{1}{2} \kappa^2 q + iq_t &= \frac{1}{2} \kappa^2 q + i\kappa\sigma_3 q_x - \frac{1}{2} q_{xx} \\ &\quad + iV^{-1}(k\sigma_3 - \alpha\sigma_1)V(-i\kappa\sigma_3 q + q_x) \\ &\quad - |q|^2 q + \frac{\Delta}{2} V^{-1} \sigma_3 V q. \end{aligned}$$

Since

$$(1.40) \quad V^{-1}(k\sigma_3 - \alpha\sigma_1)V = e^{i\kappa x \sigma_3} B^\top (k\sigma_3 - \alpha\sigma_1) B e^{-i\kappa x \sigma_3},$$

equation (1.37) implies

$$(1.41) \quad V^{-1}(k\sigma_3 - \alpha\sigma_1)V = -\kappa\sigma_3.$$

Substituting (1.41) into (1.39), the first-order derivative terms cancel:

$$(1.42) \quad iq_t = -\frac{1}{2} q_{xx} - |q|^2 q + \frac{\Delta}{2} V^{-1} \sigma_3 V q.$$

Finally,

$$(1.43) \quad \begin{aligned} V^{-1} \sigma_3 V &= e^{i\kappa x \sigma_3} B^\top \sigma_3 B e^{-i\kappa x \sigma_3} \\ &= e^{i\kappa x \sigma_3} \frac{-1}{\sqrt{\alpha^2 + k^2}} \begin{pmatrix} k & \alpha \\ \alpha & -k \end{pmatrix} e^{-i\kappa x \sigma_3}. \end{aligned}$$

Therefore,

$$(1.44) \quad iq_t = -\frac{1}{2} q_{xx} - |q|^2 q + \frac{\Delta}{2} e^{i\kappa x \sigma_3} \frac{-1}{\sqrt{\alpha^2 + k^2}} \begin{pmatrix} k & \alpha \\ \alpha & -k \end{pmatrix} e^{-i\kappa x \sigma_3} q.$$

HW 2: The squared eigenfunctions on Weierstrass points

2026-04-29

Let $B \in \mathbb{C}^{g \times g}$ be a symmetric matrix with positive definite imaginary part. The associated Riemann theta function is defined by

$$(2.1) \quad \Theta(z, B) = \sum_{n \in \mathbb{Z}^g} \exp\left(\pi i n^\top B n + 2\pi i n^\top z\right), \quad z \in \mathbb{C}^g.$$

For $k = 1, \dots, g$, let

$$e_k = (0, \dots, 0, 1, 0, \dots, 0)^\top$$

denote the k -th standard basis vector of $\mathbb{R}^g$. The theta function satisfies the quasi-periodicity relations

$$(2.2) \quad \Theta(z + e_k, B) = \Theta(z, B),$$

and

$$(2.3) \quad \Theta(z + B e_k, B) = \exp(-2\pi i z_k - \pi i B_{kk}) \Theta(z, B),$$

where z_k denotes the k -th component of z .

Consider the genus-two hyperelliptic curve

$$(2.4) \quad \mathcal{F}_2 = \left\{ P = (\lambda, y) \mid y^2 = \prod_{j=1}^3 (\lambda - \lambda_j)(\lambda - \lambda_j^*) \right\},$$

with $\lambda \in \mathbb{CP}^1$.

In Case 1, we assume that

$$\lambda_1, \lambda_2, \lambda_3 \in i\mathbb{R}, \quad \Im \lambda_3 > \Im \lambda_2 > \Im \lambda_1 > 0.$$

Let $\{a_1, a_2, b_1, b_2\}$ be a canonical homology basis satisfying

$$a_i \circ b_j = -b_j \circ a_i = \delta_{ij}, \quad a_i \circ a_j = 0, \quad b_i \circ b_j = 0.$$

Let $\{w_1 d\lambda, w_2 d\lambda\}$ be the normalized basis of holomorphic differentials determined by

$$(2.5) \quad \oint_{a_j} w_i d\lambda = \delta_{ij}, \quad \oint_{b_j} w_i d\lambda = B_{ij},$$

for $i, j = 1, 2$.

For a fixed base point $P_0 \in \mathcal{F}_2$, the Abel map is defined by

$$(2.6) \quad \mathcal{A}_{P_0}(P) = (\mathcal{A}_{P_0,1}(P), \mathcal{A}_{P_0,2}(P))^\top = \left(\int_{P_0}^P w_1 d\lambda, \int_{P_0}^P w_2 d\lambda \right)^\top.$$

Let $d\Omega_1, d\Omega_2, d\Omega_3$ be normalized Abelian differentials satisfying

$$(2.7) \quad \oint_{a_j} d\Omega_k = 0, \quad j = 1, 2, \quad k = 1, 2, 3.$$

Their corresponding Abelian integrals admit the asymptotic expansions

$$(2.8) \quad \Omega_1(P) = \pm (\ln \lambda + \ln \omega_1 + o(1)), \quad P \rightarrow \infty^\pm,$$

$$(2.9) \quad \Omega_2(P) = \pm (\lambda + \omega_2 + o(1)), \quad P \rightarrow \infty^\pm,$$

and

$$(2.10) \quad \Omega_3(P) = \pm (2\lambda^2 + \omega_3 + o(1)), \quad P \rightarrow \infty^\pm.$$

Fix the base point $P_0 = (\lambda_3, 0)$, and define

$$(2.11) \quad \Omega_k(P) = \int_{P_0}^P d\Omega_k, \quad k = 1, 2, 3.$$

The corresponding b -period vectors are defined by

$$(2.12) \quad U_k = \frac{1}{2\pi i} \oint_{b_k} d\Omega_2, \quad V_k = \frac{1}{2\pi i} \oint_{b_k} d\Omega_3, \quad \Delta_k = -\frac{1}{2\pi i} \oint_{b_k} d\Omega_1,$$

for $k = 1, 2$. If ℓ is a positively oriented circle surrounding ∞^+ , then

$$(2.13) \quad \oint_{\ell} d\Omega_1 = 2\pi i.$$

The vectors $U = (U_1, U_2)$ and $V = (V_1, V_2)$ satisfy

$$(2.14) \quad U_1 = U_2 \in i\mathbb{R},$$

and

$$(2.15) \quad V_1 = -V_2 \in \mathbb{R}.$$

Moreover,

$$(2.16) \quad \Delta_k \in \mathbb{R}, \quad \Delta_1 + \Delta_2 = 1,$$

and the symmetry relation

$$(2.17) \quad \sigma_1 \Delta = -\Delta + \mathbf{1}, \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

holds.

Define

$$(2.18) \quad \Omega = D + iUx + iVt = iU(x - x_0) + iV(t - t_0),$$

where $D = -iUx_0 - iVt_0$, and let

$$(2.19) \quad \Delta = \mathcal{A}_{P_0}(\infty^+) - \mathcal{A}_{P_0}(\infty^-) = \mathcal{A}_{\infty^-}(\infty^+).$$

Then

$$(2.20) \quad \sigma_1 \Omega^* = \Omega, \quad \mathcal{A}_{\infty^-}(P_0) = \frac{\Delta}{2}, \quad \mathcal{A}_{\infty^+}(P_0) = -\frac{\Delta}{2}.$$

The Baker–Akhiezer function associated with the two-phase solution is given by

$$(2.21) \quad \eta(P; x, t) = \omega_2 x + \omega_3 t + \frac{\theta}{2} + \frac{i}{2} \Omega_1(P),$$

and

$$(2.22) \quad \begin{pmatrix} \Phi_{11}(P; x, t) \\ \Phi_{21}(P; x, t) \end{pmatrix} = \frac{\exp(i\Omega_2(P)x + i\Omega_3(P)t)}{\Theta(\Omega)} \exp(-i\eta(P; x, t)\sigma_3) \\ \times \begin{pmatrix} \Theta(\Omega + \mathcal{A}_{\infty^+}(P)) \\ \Theta(\Omega + \Delta + \mathcal{A}_{\infty^+}(P)) \end{pmatrix}.$$

Here $\omega_2 = 0$ and $\omega_3 \in \mathbb{R}$.

Using (2.20), we obtain

$$(2.23) \quad \Omega + \mathcal{A}_{\infty^+}(P) = \Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P),$$

and

$$(2.24) \quad \Omega + \Delta + \mathcal{A}_{\infty^+}(P) = \Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P).$$

Define the branch points

$$(2.25) \quad P_j = (\lambda_j, 0), \quad P_{-j} = (\lambda_j^*, 0), \quad j = 1, 2, 3.$$

For $P \in \mathcal{F}_2$, define

$$(2.26) \quad P_1(P) = \begin{pmatrix} \Phi_{11}(P) \\ \Phi_{21}(P) \end{pmatrix} (\Phi_{11}^*(\tau_R(\iota(P))), \Phi_{21}^*(\tau_R(\iota(P)))) ,$$

and

$$(2.27) \quad S_1(P) = \begin{pmatrix} \Phi_{21}(P) \Phi_{11}^*(\tau_R(\iota(P))) \\ \Phi_{11}(P) \Phi_{21}^*(\tau_R(\iota(P))) \end{pmatrix}.$$

Here τ_R and ι denote the involutions

$$(2.28) \quad \tau_R : (\lambda, y) \mapsto (\lambda^*, y^*), \quad \iota : (\lambda, y) \mapsto (\lambda, -y).$$

Let

$$(2.29) \quad \mathcal{P} : \mathcal{F}_2 \rightarrow \mathbb{CP}^1, \quad \mathcal{P}(\lambda, y) = \lambda,$$

be the canonical projection map. We consider $S_1(P)$ at the Weierstrass points satisfying

$$\mathcal{P}(P) = \pm\lambda_1, \pm\lambda_2, \pm\lambda_3.$$

Since P is a branch point, one has $\iota(P) = P$. Hence

$$(2.30) \quad S_1(P) = \begin{pmatrix} \Phi_{21}(P) \Phi_{11}^*(\tau_R(P)) \\ \Phi_{11}(P) \Phi_{21}^*(\tau_R(P)) \end{pmatrix}.$$

We now establish the following result concerning the squared eigenfunctions evaluated at the Weierstrass points.

Proposition 1. *The vectors*

$$S_1(P_j), \quad j = \pm 1, \pm 2, \pm 3,$$

span a three-dimensional subspace. More precisely,

$$(2.31) \quad \dim(\text{span}\{S_1(P_j) : j = \pm 1, \pm 2, \pm 3\}) = 3.$$

Define

$$(2.32) \quad \begin{pmatrix} \tilde{\Phi}_{11}(P) \\ \tilde{\Phi}_{21}(P) \end{pmatrix} = \exp \left[-i \left(\omega_2 x + \omega_3 t + \frac{\theta}{2} + \frac{i}{2} \Omega_1(P) \right) \sigma_3 \right] \begin{pmatrix} \Theta \left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P) \right) \\ \Theta \left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P) \right) \end{pmatrix}.$$

Taking complex conjugates at the reflected point $\tau_R(P)$, we obtain

$$(2.33) \quad \begin{aligned} & \begin{pmatrix} \tilde{\Phi}_{11}^*(\tau_R(P)) \\ \tilde{\Phi}_{21}^*(\tau_R(P)) \end{pmatrix} \\ &= \begin{pmatrix} \Theta^* \left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(\tau_R(P)) \right) \\ \Theta^* \left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(\tau_R(P)) \right) \end{pmatrix} \\ &\times \exp \left[i \left(\omega_2 x + \omega_3 t + \frac{\theta}{2} - \frac{i}{2} \Omega_1^*(\tau_R(P)) \right) \sigma_3 \right]. \end{aligned}$$

Substituting these expressions into the definition of $S_1(P)$ yields

$$(2.34) \quad S_1(P) = \tilde{S}_1(P) \frac{\exp(i(\Omega_2(P) - \Omega_2^*(\tau_R(P)))x + i(\Omega_3(P) - \Omega_3^*(\tau_R(P)))t)}{|\Theta(\Omega)|^2}.$$

Here

$$(2.35) \quad \tilde{S}_1(P) = \begin{pmatrix} \tilde{\Phi}_{21}(P) \tilde{\Phi}_{11}^*(\tau_R(P)) \\ \tilde{\Phi}_{11}(P) \tilde{\Phi}_{21}^*(\tau_R(P)) \end{pmatrix}.$$

Using (2.32) and (2.33), we further obtain

$$(2.36) \quad \begin{aligned} \tilde{S}_1(P) &= \exp \left[i \left(2\omega_2 x + 2\omega_3 t + \theta + \frac{i}{2} (\Omega_1(P) - \Omega_1^*(\tau_R(P))) \right) \sigma_3 \right] \\ &\times \begin{pmatrix} \Theta \left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P) \right) \Theta^* \left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(\tau_R(P)) \right) \\ \Theta \left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P) \right) \Theta^* \left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(\tau_R(P)) \right) \end{pmatrix}. \end{aligned}$$

Since

$$(2.37) \quad \mathcal{A}_{P_0}(P) = \begin{pmatrix} \int_{P_0}^P w_1 d\lambda \\ \int_{P_0}^P w_2 d\lambda \end{pmatrix}, \quad P_0 = P_3 = (\lambda_3, 0),$$

we choose all integration paths in the sheet $\mathcal{F}^-$. Then

$$(2.38) \quad \mathcal{A}_{P_0}(P_3) = 0.$$

For the branch point P_2 , the integration path corresponds to one half of the cycle b_1 . Hence

$$(2.39) \quad \mathcal{A}_{P_0}(P_2) = \frac{1}{2} \oint_{b_1} \begin{pmatrix} w_1 d\lambda \\ w_2 d\lambda \end{pmatrix} = \frac{B\mathbf{e}_1}{2},$$

where

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Similarly, the path from P_3 to P_1 is homologous to one half of the cycle $-a_1 + b_1$, and therefore

$$(2.40) \quad \mathcal{A}_{P_0}(P_1) = \frac{1}{2} \oint_{-a_1+b_1} \begin{pmatrix} w_1 d\lambda \\ w_2 d\lambda \end{pmatrix} = \frac{B\mathbf{e}_1 - \mathbf{e}_1}{2}.$$

For the conjugate branch point P_{-3} , the integration path is homologous to one half of the cycle $-a_1 - a_2$. Thus

$$(2.41) \quad \mathcal{A}_{P_0}(P_{-3}) = \frac{1}{2} \oint_{-a_1-a_2} \begin{pmatrix} w_1 d\lambda \\ w_2 d\lambda \end{pmatrix} = -\frac{\mathbf{e}_1 + \mathbf{e}_2}{2} = -\frac{\mathbf{1}}{2},$$

where

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Proceeding in the same way, we obtain

$$(2.42) \quad \mathcal{A}_{P_0}(P_{-2}) = \frac{1}{2} \oint_{b_2-a_1-a_2} \begin{pmatrix} w_1 d\lambda \\ w_2 d\lambda \end{pmatrix} = \frac{B\mathbf{e}_2 - \mathbf{1}}{2},$$

and

$$(2.43) \quad \mathcal{A}_{P_0}(P_{-1}) = \frac{1}{2} \oint_{b_2-a_1} \begin{pmatrix} w_1 d\lambda \\ w_2 d\lambda \end{pmatrix} = \frac{B\mathbf{e}_2 - \mathbf{e}_1}{2}.$$

Since $P_0 = P_3$, we immediately obtain

$$(2.44) \quad \Omega_1(P_3) = \int_{P_0}^{P_3} d\Omega_1 = 0.$$

Let ℓ be a positively oriented small circle surrounding ∞^+ . Since

$$\oint_{a_j} d\Omega_1 = 0,$$

the integration path from $P_0 = P_3$ to P_{-3} is homologous to one half of the cycle ℓ . Hence

$$(2.45) \quad \Omega_1(P_{-3}) = \frac{1}{2} \oint_{\ell} d\Omega_1 = \frac{1}{2} \cdot 2\pi i = \pi i.$$

Similarly, the integration paths corresponding to P_1 and P_2 are homologous to one half of the cycle $\ell + b_1$. Hence

$$(2.46) \quad \Omega_1(P_1) = \Omega_1(P_2) = \frac{1}{2} \oint_{\ell+b_1} d\Omega_1 = \pi i + \frac{1}{2} \oint_{b_1} d\Omega_1 = \pi i - \pi i \Delta_1 = \pi i(1 - \Delta_1).$$

In the same way,

$$(2.47) \quad \Omega_1(P_{-1}) = \Omega_1(P_{-2}) = \frac{1}{2} \oint_{\ell+b_2} d\Omega_1 = \pi i + \frac{1}{2} \oint_{b_2} d\Omega_1 = \pi i - \pi i \Delta_2 = \pi i(1 - \Delta_2).$$

Since

$$\Omega_1(P_j) \in i\mathbb{R}, \quad \Delta_1 + \Delta_2 = 1,$$

we obtain

$$(2.48) \quad \Omega_1(P_j) - \Omega_1^*(\tau_R(P_j)) = \Omega_1(P_j) + \Omega_1(P_{-j}) = \pi i, \quad j = \pm 1, \pm 2, \pm 3.$$

Consequently,

$$(2.49) \quad \frac{i}{2} \left(\Omega_1(P_j) - \Omega_1^*(\tau_R(P_j)) \right) = \frac{i}{2} \pi i = -\frac{\pi}{2}.$$

By analogous arguments, one also obtains

$$(2.50) \quad \Omega_2(P_j) - \Omega_2^*(\tau_R(P_j)) = 0, \quad \Omega_3(P_j) - \Omega_3^*(\tau_R(P_j)) = 0, \quad j = \pm 1, \pm 2, \pm 3.$$

Substituting these identities into (2.34), we conclude that

$$(2.51) \quad S_1(P_j) = \frac{\tilde{S}_1(P_j)}{|\Theta(\Omega)|^2}, \quad j = \pm 1, \pm 2, \pm 3.$$

Define

$$(2.52) \quad \tilde{\theta}_1 = 2\omega_2 x + 2\omega_3 t + \theta - \frac{\pi}{2}.$$

Then (2.36) becomes

$$(2.53) \quad \tilde{S}_1(P_j) = \exp(i\tilde{\theta}_1 \sigma_3) \begin{pmatrix} \Theta\left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\right) \Theta^*\left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_{-j})\right) \\ \Theta\left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\right) \Theta^*\left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_{-j})\right) \end{pmatrix}.$$

Using the identities

$$\Theta^*(z) = \Theta(z^*), \quad \Theta(z) = \Theta(\sigma_1 z), \quad \sigma_1 \Omega^* = \Omega,$$

we may rewrite (2.53) as

$$(2.54) \quad \tilde{S}_1(P_j) = \exp(i\tilde{\theta}_1 \sigma_3) \begin{pmatrix} \Theta\left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\right) \Theta\left(\Omega - \frac{\sigma_1 \Delta}{2} + \sigma_1 \mathcal{A}_{P_0}^*(P_{-j})\right) \\ \Theta\left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\right) \Theta\left(\Omega + \frac{\sigma_1 \Delta}{2} + \sigma_1 \mathcal{A}_{P_0}^*(P_{-j})\right) \end{pmatrix}.$$

Since

$$\sigma_1 \Delta = -\Delta + \mathbf{1},$$

we further obtain

$$(2.55) \quad \tilde{S}_1(P_j) = \exp(i\tilde{\theta}_1 \sigma_3) \begin{pmatrix} \Theta\left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\right) \Theta\left(\Omega + \frac{\Delta}{2} - \frac{\mathbf{1}}{2} + \sigma_1 \mathcal{A}_{P_0}^*(P_{-j})\right) \\ \Theta\left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\right) \Theta\left(\Omega - \frac{\Delta}{2} + \frac{\mathbf{1}}{2} + \sigma_1 \mathcal{A}_{P_0}^*(P_{-j})\right) \end{pmatrix}.$$

Finally, using the periodicity relation

$$\Theta(z + e_k) = \Theta(z), \quad k = 1, 2,$$

we conclude from (2.55) that

$$(2.56) \quad \tilde{S}_1(P_j) = \exp(i\tilde{\theta}_1 \sigma_3) \begin{pmatrix} \Theta\left(\Omega + \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\right) \Theta\left(\Omega + \frac{\Delta}{2} + \frac{\mathbf{1}}{2} + \sigma_1 \mathcal{A}_{P_0}^*(P_{-j})\right) \\ \Theta\left(\Omega - \frac{\Delta}{2} + \mathcal{A}_{P_0}(P_j)\right) \Theta\left(\Omega - \frac{\Delta}{2} + \frac{\mathbf{1}}{2} + \sigma_1 \mathcal{A}_{P_0}^*(P_{-j})\right) \end{pmatrix}.$$

Note that

$$(2.57) \quad \mathcal{A}_{P_0}(P_j) + \sigma_1 \mathcal{A}_{P_0}^*(P_{-j}) + \frac{\mathbf{1}}{2} = \mathbf{0}, \quad j = \pm 1, \pm 2, \pm 3.$$

For example, using (2.39) and (2.42), we obtain

$$(2.58) \quad \mathcal{A}_{P_0}(P_2) + \sigma_1 \mathcal{A}_{P_0}^*(P_{-2}) + \frac{1}{2} = \frac{Be_1}{2} + \sigma_1 \frac{-Be_2 - 1}{2} + \frac{1}{2} = \frac{Be_1 - \sigma_1 Be_2}{2} = 0.$$

Similarly,

$$(2.59) \quad \mathcal{A}_{P_0}(P_{-2}) + \sigma_1 \mathcal{A}_{P_0}^*(P_2) + \frac{1}{2} = \frac{Be_2 - 1}{2} + \sigma_1 \frac{-Be_1}{2} + \frac{1}{2} = \frac{Be_2 - \sigma_1 Be_1}{2} = 0.$$

Here we have used the symmetry relations

$$(2.60) \quad Be_1 = \sigma_1 Be_2, \quad B_{ij} \in i\mathbb{R}.$$

Introduce the notation

$$(2.61) \quad \Theta_n^m(z, B) = \begin{pmatrix} \Theta\left(z + \frac{\Delta}{2} + \frac{n+Bm}{2}, B\right) \Theta\left(z + \frac{\Delta}{2} - \frac{n+Bm}{2}, B\right) \\ \Theta\left(z - \frac{\Delta}{2} + \frac{n+Bm}{2}, B\right) \Theta\left(z - \frac{\Delta}{2} - \frac{n+Bm}{2}, B\right) \end{pmatrix}.$$

Using (2.56) together with (2.57), we first obtain

$$(2.62) \quad \tilde{S}_1(P_3) = \exp(i\tilde{\theta}_1\sigma_3) \begin{pmatrix} \Theta\left(\Omega + \frac{\Delta}{2}\right)^2 \\ \Theta\left(\Omega - \frac{\Delta}{2}\right)^2 \end{pmatrix} = \exp(i\tilde{\theta}_1\sigma_3) \Theta_0^0(\Omega, B).$$

For P_{-3} , using (2.41), we have

$$(2.63) \quad \begin{aligned} \tilde{S}_1(P_{-3}) &= \exp(i\tilde{\theta}_1\sigma_3) \begin{pmatrix} \Theta\left(\Omega + \frac{\Delta}{2} - \frac{1}{2}\right) \Theta\left(\Omega + \frac{\Delta}{2} + \frac{1}{2}\right) \\ \Theta\left(\Omega - \frac{\Delta}{2} - \frac{1}{2}\right) \Theta\left(\Omega - \frac{\Delta}{2} + \frac{1}{2}\right) \end{pmatrix} \\ &= \exp(i\tilde{\theta}_1\sigma_3) \Theta_1^0(\Omega, B). \end{aligned}$$

Similarly,

$$(2.64) \quad \begin{aligned} \tilde{S}_1(P_2) &= \exp(i\tilde{\theta}_1\sigma_3) \begin{pmatrix} \Theta\left(\Omega + \frac{\Delta}{2} + \frac{Be_1}{2}\right) \Theta\left(\Omega + \frac{\Delta}{2} - \frac{Be_1}{2}\right) \\ \Theta\left(\Omega - \frac{\Delta}{2} + \frac{Be_1}{2}\right) \Theta\left(\Omega - \frac{\Delta}{2} - \frac{Be_1}{2}\right) \end{pmatrix} \\ &= \exp(i\tilde{\theta}_1\sigma_3) \Theta_0^{\epsilon_1}(\Omega, B). \end{aligned}$$

In the same way,

$$(2.65) \quad \tilde{S}_1(P_{-2}) = \exp(i\tilde{\theta}_1\sigma_3) \Theta_1^{\epsilon_2}(\Omega, B),$$

and

$$(2.66) \quad \tilde{S}_1(P_1) = \exp(i\tilde{\theta}_1\sigma_3) \Theta_{\epsilon_1}^{\epsilon_1}(\Omega, B), \quad \tilde{S}_1(P_{-1}) = \exp(i\tilde{\theta}_1\sigma_3) \Theta_{\epsilon_1}^{\epsilon_2}(\Omega, B).$$

The Riemann addition formula states that

$$(2.67) \quad \Theta(z + w, B) \Theta(z - w, B) = \sum_{\epsilon \in \{0,1\}^2} \Theta \begin{bmatrix} \epsilon \\ 0 \end{bmatrix} (2w, 2B) \Theta \begin{bmatrix} \epsilon \\ 0 \end{bmatrix} (2z, 2B).$$

Here the theta function with characteristics is defined by

$$(2.68) \quad \Theta \begin{bmatrix} \epsilon \\ \epsilon' \end{bmatrix} (z, B) = \exp \left(2\pi i \left(\frac{\epsilon^\top B \epsilon}{8} + \frac{\epsilon^\top z}{2} + \frac{\epsilon^\top \epsilon'}{4} \right) \right) \Theta \left(z + \frac{\epsilon' + B\epsilon}{2}, B \right),$$

and

$$(2.69) \quad \{0, 1\}^2 = \{\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{1}\}.$$

Let

$$(2.70) \quad w = \frac{n + B\mathbf{m}}{2}.$$

Applying (2.67) to each component of $\Theta_n^m(z, B)$, we obtain

$$(2.71) \quad \Theta_n^m(z, B) = \sum_{\varepsilon \in \{0,1\}^2} \Theta \begin{bmatrix} \varepsilon \\ \mathbf{0} \end{bmatrix} (n + B\mathbf{m}, 2B) \begin{pmatrix} \Theta \begin{bmatrix} \varepsilon \\ \mathbf{0} \end{bmatrix} (2z + \Delta, 2B) \\ \Theta \begin{bmatrix} \varepsilon \\ \mathbf{0} \end{bmatrix} (2z - \Delta, 2B) \end{pmatrix}.$$

Define

$$(2.72) \quad f_j(x, t) = f_\varepsilon(\Omega) = \begin{pmatrix} \Theta \begin{bmatrix} \varepsilon \\ \mathbf{0} \end{bmatrix} (2\Omega + \Delta, 2B) \\ \Theta \begin{bmatrix} \varepsilon \\ \mathbf{0} \end{bmatrix} (2\Omega - \Delta, 2B) \end{pmatrix}, \quad \varepsilon \in \{\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{1}\},$$

where $j = 1, 2, 3, 4$ corresponds respectively to $\varepsilon = \mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{1}$.

Next, define

$$(2.73) \quad \tilde{a}_{jk} = \tilde{a}_{\varepsilon, (m, n)} = \Theta \begin{bmatrix} \varepsilon \\ \mathbf{0} \end{bmatrix} (n + B\mathbf{m}, 2B),$$

where

$$\varepsilon = \mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{1}, \quad k = 1, 2, \dots, 6,$$

and

$$(2.74) \quad \begin{pmatrix} \mathbf{m} \\ \mathbf{n} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \quad \begin{pmatrix} \mathbf{0} \\ \mathbf{1} \end{pmatrix}, \quad \begin{pmatrix} \mathbf{e}_1 \\ \mathbf{0} \end{pmatrix}, \quad \begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_1 \end{pmatrix}, \quad \begin{pmatrix} \mathbf{e}_2 \\ \mathbf{e}_1 \end{pmatrix}, \quad \begin{pmatrix} \mathbf{e}_2 \\ \mathbf{1} \end{pmatrix}.$$

For $k = 1, \dots, 6$, define

$$(2.75) \quad g_k(x, t) = \exp(-i\tilde{\theta}_1\sigma_3) \tilde{S}_1(P_j), \quad j = 3, -3, 2, 1, -1, -2,$$

respectively. Equivalently,

$$g_1 = \exp(-i\tilde{\theta}_1\sigma_3) \tilde{S}_1(P_3), \quad g_2 = \exp(-i\tilde{\theta}_1\sigma_3) \tilde{S}_1(P_{-3}), \quad \dots, \quad g_6 = \exp(-i\tilde{\theta}_1\sigma_3) \tilde{S}_1(P_{-2}).$$

Then (2.56) and (2.71) imply that

$$(2.76) \quad g_k = \sum_{j=1}^4 \tilde{a}_{jk} f_j, \quad k = 1, 2, \dots, 6.$$

Equivalently,

$$(2.77) \quad [g_1, g_2, \dots, g_6] = [f_1, f_2, f_3, f_4] \tilde{A}, \quad \tilde{A}_{jk} = \tilde{a}_{jk}.$$

Since the functions f_j are linearly independent and $\exp(i\tilde{\theta}_1\sigma_3)$ is an invertible diagonal matrix, multiplying by this factor does not affect linear dependence. Hence it suffices to determine $\text{rank}(\tilde{A})$.

Using (2.68), we compute

$$(2.78) \quad \begin{aligned} \Theta \begin{bmatrix} \varepsilon \\ \mathbf{0} \end{bmatrix} (n + B\mathbf{m}, 2B) &= \exp \left(2\pi i \left(\frac{\varepsilon^\top B\varepsilon}{4} + \frac{\varepsilon^\top (n + B\mathbf{m})}{2} \right) \right) \Theta(n + B\mathbf{m} + B\varepsilon, 2B) \\ &= \exp \left(\frac{\pi i}{2} \varepsilon^\top B\varepsilon \right) \exp \left(\pi i \varepsilon^\top (n + B\mathbf{m}) \right) \Theta(B(\mathbf{m} + \varepsilon), 2B). \end{aligned}$$

Motivated by this expression, define

$$(2.79) \quad a_{jk} = a_{\varepsilon, (m, n)} = \exp \left(\pi i \varepsilon^\top (n + B\mathbf{m}) \right) \Theta(B(\mathbf{m} + \varepsilon), 2B),$$

for $j = 1, \dots, 4$ and $k = 1, \dots, 6$. Then

$$(2.80) \quad \tilde{A} = \text{diag} \left(1, \exp \left(\frac{\pi i}{2} \mathbf{e}_1^\top B \mathbf{e}_1 \right), \exp \left(\frac{\pi i}{2} \mathbf{e}_2^\top B \mathbf{e}_2 \right), \exp \left(\frac{\pi i}{2} \mathbf{1}^\top B \mathbf{1} \right) \right) A,$$

where $A_{jk} = a_{jk}$.

Denote

$$(2.81) \quad \Theta_{ij} = \Theta(B(i\mathbf{e}_1 + j\mathbf{e}_2), 2B), \quad i, j \in \{0, 1\}.$$

From (2.79), we first obtain

$$(2.82) \quad a_{j1} = \Theta(B\boldsymbol{\varepsilon}, 2B), \quad a_{j2} = \exp(\pi i \boldsymbol{\varepsilon}^\top \mathbf{1}) \Theta(B\boldsymbol{\varepsilon}, 2B).$$

Similarly,

$$(2.83) \quad a_{j3} = \exp(\pi i \boldsymbol{\varepsilon}^\top B \mathbf{e}_1) \Theta(B(\mathbf{e}_1 + \boldsymbol{\varepsilon}), 2B),$$

and

$$(2.84) \quad a_{j4} = \exp(\pi i \boldsymbol{\varepsilon}^\top (\mathbf{e}_1 + B \mathbf{e}_1)) \Theta(B(\mathbf{e}_1 + \boldsymbol{\varepsilon}), 2B).$$

Moreover,

$$(2.85) \quad a_{j5} = \exp(\pi i \boldsymbol{\varepsilon}^\top (\mathbf{e}_1 + B \mathbf{e}_2)) \Theta(B(\mathbf{e}_2 + \boldsymbol{\varepsilon}), 2B),$$

and

$$(2.86) \quad a_{j6} = \exp(\pi i \boldsymbol{\varepsilon}^\top (\mathbf{1} + B \mathbf{e}_2)) \Theta(B(\mathbf{e}_2 + \boldsymbol{\varepsilon}), 2B).$$

Using the ordering

$$\boldsymbol{\varepsilon} = \mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{1},$$

the first two column vectors of A are

$$(2.87) \quad A_1 = \begin{pmatrix} \Theta_{00} \\ \Theta_{10} \\ \Theta_{01} \\ \Theta_{11} \end{pmatrix}, \quad A_2 = \begin{pmatrix} \Theta_{00} \\ -\Theta_{10} \\ -\Theta_{01} \\ \Theta_{11} \end{pmatrix}.$$

For the third column, we compute

$$(2.88) \quad A_3 = \begin{pmatrix} \Theta_{10} \\ e^{\pi i B_{11}} e^{-2\pi i B_{11}} \Theta_{00} \\ e^{\pi i B_{21}} \Theta_{11} \\ e^{\pi i (B_{11} + B_{21})} e^{-2\pi i (B_{11} + B_{21})} \Theta_{01} \end{pmatrix} = \begin{pmatrix} \Theta_{10} \\ e^{-\pi i B_{11}} \Theta_{00} \\ e^{\pi i B_{21}} \Theta_{11} \\ e^{-\pi i (B_{11} + B_{21})} \Theta_{01} \end{pmatrix}.$$

Similarly,

$$(2.89) \quad A_4 = \begin{pmatrix} \Theta_{10} \\ -e^{-\pi i B_{11}} \Theta_{00} \\ e^{\pi i B_{21}} \Theta_{11} \\ -e^{-\pi i (B_{11} + B_{21})} \Theta_{01} \end{pmatrix}.$$

Proceeding in the same way, we obtain

$$(2.90) \quad A_5 = \begin{pmatrix} \Theta_{01} \\ -e^{\pi i B_{21}} \Theta_{11} \\ e^{\pi i B_{22}} e^{-2\pi i B_{22}} \Theta_{00} \\ -e^{\pi i (B_{12} + B_{22})} e^{-2\pi i (B_{12} + B_{22})} \Theta_{10} \end{pmatrix} = \begin{pmatrix} \Theta_{01} \\ -e^{\pi i B_{21}} \Theta_{11} \\ e^{-\pi i B_{11}} \Theta_{00} \\ -e^{-\pi i (B_{21} + B_{11})} \Theta_{10} \end{pmatrix}.$$

Likewise,

$$(2.91) \quad A_6 = \begin{pmatrix} \Theta_{01} \\ -e^{\pi i B_{21}} \Theta_{11} \\ -e^{-\pi i B_{11}} \Theta_{00} \\ e^{-\pi i (B_{21} + B_{11})} \Theta_{10} \end{pmatrix}.$$

Hence the matrix A takes the form

$$(2.92) \quad A = [A_1, A_2, \dots, A_6].$$

Applying the column transformations

$$A \longrightarrow \left[\frac{A_1 + A_2}{2}, \frac{A_1 - A_2}{2}, \frac{A_3 + A_4}{2}, \frac{A_3 - A_4}{2}, \frac{A_5 + A_6}{2}, \frac{A_5 - A_6}{2} \right],$$

the matrix A is transformed into

$$(2.93) \quad \begin{pmatrix} \Theta_{00} & 0 & \Theta_{10} & 0 & \Theta_{01} & 0 \\ 0 & \Theta_{10} & 0 & e^{-\pi i B_{11}} \Theta_{00} & -e^{\pi i B_{21}} \Theta_{11} & 0 \\ 0 & \Theta_{01} & e^{\pi i B_{21}} \Theta_{11} & 0 & 0 & e^{-\pi i B_{11}} \Theta_{00} \\ \Theta_{11} & 0 & 0 & e^{-\pi i (B_{11} + B_{21})} \Theta_{01} & 0 & -e^{-\pi i (B_{21} + B_{11})} \Theta_{10} \end{pmatrix}.$$

Introduce the notation

$$(2.94) \quad a = \Theta_{00}, \quad b = \Theta_{10}, \quad c = \Theta_{01}, \quad d = \Theta_{11}.$$

Then (2.93) becomes

$$(2.95) \quad \begin{pmatrix} a & 0 & b & 0 & c & 0 \\ 0 & b & 0 & e^{-\pi i B_{11}} a & -e^{\pi i B_{21}} d & 0 \\ 0 & c & e^{\pi i B_{21}} d & 0 & 0 & e^{-\pi i B_{11}} a \\ d & 0 & 0 & e^{-\pi i (B_{11} + B_{21})} c & 0 & -e^{-\pi i (B_{21} + B_{11})} b \end{pmatrix}.$$

We now perform elementary row operations. First, eliminate the entry d in the fourth row and first column by replacing

$$R_4 \longrightarrow R_4 - \frac{d}{a} R_1.$$

Next, eliminate the entry c in the third row and second column by replacing

$$R_3 \longrightarrow R_3 - \frac{c}{b} R_2.$$

Finally, eliminate the remaining nonzero entries in the fourth row using the second and third rows. This yields

$$(2.96) \quad \begin{pmatrix} a & 0 & b & 0 & c & 0 \\ 0 & b & 0 & e^{-\pi i B_{11}} a & -e^{\pi i B_{21}} d & 0 \\ 0 & 0 & e^{\pi i B_{21}} d & -e^{-\pi i B_{11}} \frac{ac}{b} & e^{\pi i B_{21}} \frac{cd}{b} & e^{-\pi i B_{11}} a \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Therefore, $\text{rank}(A) = 3$. Since $\tilde{A}$ differs from A only by multiplication with an invertible diagonal matrix, one also has $\text{rank}(\tilde{A}) = 3$. Therefore Proposition 1 follows:

$$(2.97) \quad \dim(\text{span}\{S_1(P_j) : j = \pm 1, \pm 2, \pm 3\}) = \dim(\text{span}\{g_k : k = 1, 2, \dots, 6\}) = 3.$$

HW 3: Numerical Reproduction of One-Mode Anomalous-Wave Recurrence 2026-05-20

In this homework we reproduce the first numerical figure in the paper of P. G. Grinevich and P. M. Santini, *The finite-gap method and the periodic NLS Cauchy problem of anomalous waves for a finite number of unstable modes*, Russian Mathematical Surveys **74** (2019), 211–263. The main purpose is not only to draw a similar picture, but also to explain how the numerical simulation is built from the equation. I write the explanation in a beginner-oriented way: first the continuous model, then the discrete grid, then the time stepping formula, and finally the MATLAB implementation.

The equation considered in the paper is the focusing nonlinear Schrödinger equation

$$(3.1) \quad iu_t + u_{xx} + 2|u|^2u = 0, \quad u = u(x, t) \in \mathbb{C},$$

with periodic boundary conditions in x . Here x is the spatial variable, t is time, and the unknown $u(x, t)$ is complex-valued. The quantity that we plot is the amplitude $|u(x, t)|$. The exact constant-amplitude background is

$$(3.2) \quad u_0(x, t) = e^{2it},$$

so $|u_0(x, t)| = 1$ for all x and t . The numerical experiment starts from a very small periodic perturbation of the unit background at $t = 0$, and the interesting question is whether this tiny perturbation stays tiny or grows into a large visible wave.

Modulation instability. Modulation instability is the growth of a small long-wave perturbation of a plane wave background. For the focusing NLS equation (3.1), write the solution near the background in the form

$$(3.3) \quad u(x, t) = e^{2it} (1 + w(x, t)), \quad |w| \ll 1.$$

After linearization, each Fourier pair with wave number k evolves as a combination of a growing and a decaying mode when

$$(3.4) \quad 0 < k < 2.$$

More precisely, the growth rate is

$$(3.5) \quad \sigma(k) = k\sqrt{4 - k^2}, \quad 0 < k < 2.$$

If $k > 2$, the corresponding linearized frequency is real and the mode does not grow exponentially. This is the first important idea behind the picture: not every Fourier mode is dangerous. Only the modes whose wave numbers lie in the interval $0 < k < 2$ are linearly unstable. Thus the focusing sign in (3.1) is essential: it is the reason why a perturbation of size 10^{-4} can later become an $O(1)$ coherent wave.

On a periodic interval of length L , the admissible Fourier wave numbers are

$$(3.6) \quad k_j = \frac{2\pi j}{L}, \quad j = 1, 2, \dots$$

Therefore $k_j < 2$ is equivalent to $j < L/\pi$, and the number of unstable positive Fourier modes is

$$(3.7) \quad N = \left\lfloor \frac{L}{\pi} \right\rfloor.$$

This is the basic mechanism behind the phrase “a finite number of unstable modes” in the paper.

One unstable mode. The initial condition used for Figure 1 of the paper is

$$(3.8) \quad u(x, 0) = 1 + \epsilon \left(c_1 e^{ik_1 x} + c_{-1} e^{-ik_1 x} \right), \quad k_1 = \frac{2\pi}{L},$$

where

$$(3.9) \quad L = 6, \quad \epsilon = 10^{-4}, \quad c_1 = \frac{1}{2}, \quad c_{-1} = \frac{0.3 - 0.4i}{2}.$$

For these parameters,

$$(3.10) \quad N = \left\lfloor \frac{L}{\pi} \right\rfloor = 1, \quad k_1 = \frac{\pi}{3} < 2,$$

so only the first Fourier mode belongs to the unstable band (3.4). The second mode would have wave number $2k_1 = 2\pi/3 > 2$, so it is outside the unstable band. Thus this is a clean one-mode test case. The perturbation first grows from the almost constant background, then forms a localized Akhmediev-type peak, and finally returns close to the background before the next recurrence.

Split-step Fourier discretization. The split-step Fourier method is a standard method for equations such as (3.1), because the equation contains two parts that are easy in two different representations. First rewrite the equation as

$$(3.11) \quad u_t = iu_{xx} + 2i|u|^2u.$$

The term iu_{xx} is the linear dispersive part. It is easy after applying the Fourier transform. The term $2i|u|^2u$ is the nonlinear part. It is easy at each fixed grid point in physical space. The split-step idea is to solve these two parts one after the other during each small time interval.

The spatial interval in the code is

$$[-L/2, L/2] = [-3, 3],$$

and the code uses $N_x = 384$ grid points. The grid spacing is therefore

$$\Delta x = \frac{L}{N_x}.$$

The grid is periodic, so the left endpoint and the right endpoint represent the same point. This is why the MATLAB code creates $N_x + 1$ points by `linspace` and then removes the last one. The Fourier wave numbers used by MATLAB's FFT are ordered as

$$(3.12) \quad k = \frac{2\pi}{L} \left(0, 1, \dots, \frac{N_x}{2} - 1, -\frac{N_x}{2}, \dots, -1 \right).$$

This ordering is not arbitrary: it is the ordering expected by the vectors produced by `fft`.

For the linear subproblem

$$u_t = iu_{xx},$$

write

$$u(x, t) = \sum_k \hat{u}_k(t) e^{ikx}.$$

Since differentiating twice gives

$$u_{xx} = \sum_k (-k^2) \hat{u}_k(t) e^{ikx},$$

each Fourier coefficient satisfies

$$\frac{d}{dt} \hat{u}_k = -ik^2 \hat{u}_k.$$

Therefore the exact linear evolution over a time interval τ is

$$(3.13) \quad \hat{u}_k(t + \tau) = e^{-ik^2\tau} \hat{u}_k(t).$$

In the MATLAB code, one half linear step has $\tau = \Delta t/2$, so the multiplier is

$$E_{\text{half}} = \exp(-1i * k.^2 * dt/2).$$

For the nonlinear subproblem

$$u_t = 2i|u|^2u,$$

there is no spatial derivative. Thus each grid value evolves independently. During this nonlinear substep the modulus $|u|$ is constant, and only the complex phase rotates. The exact nonlinear evolution over one full time step Δt is

$$(3.14) \quad u(t + \Delta t) = u(t) \exp(2i|u(t)|^2 \Delta t).$$

This is exactly the MATLAB command

$$u = u .* \exp(2i * \text{abs}(u).^2 * dt).$$

The code uses $T_{\max} = 60$, $N_t = 30000$, and hence

$$(3.15) \quad \Delta t = \frac{T_{\max}}{N_t} = \frac{60}{30000} = 0.002.$$

The number 0.002 is small enough for a smooth reproduction while keeping the runtime short. One full Strang step is

$$(3.16) \quad u^{n+1} = \mathcal{L}_{\Delta t/2} \mathcal{N}_{\Delta t} \mathcal{L}_{\Delta t/2} u^n,$$

where

$$\mathcal{L}_{\Delta t/2} u = \mathcal{F}^{-1} \left[e^{-ik^2 \Delta t/2} \mathcal{F} u \right], \quad \mathcal{N}_{\Delta t} u = u \exp(2i|u|^2 \Delta t).$$

Written out in the same order as the MATLAB loop, this is

$$(3.17) \quad u^{(1)} = \mathcal{F}^{-1} \left[e^{-ik^2 \Delta t/2} \mathcal{F} u^n \right],$$

$$(3.18) \quad u^{(2)} = u^{(1)} \exp\left(2i|u^{(1)}|^2 \Delta t\right),$$

$$(3.19) \quad u^{n+1} = \mathcal{F}^{-1} \left[e^{-ik^2 \Delta t/2} \mathcal{F} u^{(2)} \right].$$

Equations (3.17)–(3.19) are the mathematical version of the three main operations inside the `for` loop: FFT half-step, nonlinear phase rotation, FFT half-step.

MATLAB code and explanation. The final script used for the paper-style black-and-white reproduction is `reproduce_paper_fig1_bw_v6.m`. It also saves and reuses `fig1_data.mat`, so the expensive time integration does not need to be repeated every time we only want to adjust the appearance of the figure.

```

1 function reproduce_paper_fig1_bw_v6()
2
3     clear; clc; close all;
4
5     %% =====
6     % Parameters from the paper
7     %% =====
8
9     L = 6;
10    eps0 = 1e-4;
11
12    c1 = 1/2;
13    cm1 = (0.3 - 0.4i)/2;
14
15    k1 = 2*pi/L;
16
17    %% =====
18    % Numerical parameters
19    %% =====
20
21    Nx = 384;
22    Tmax = 60;
23
24    Nt = 30000;
25    dt = Tmax/Nt;
26
27    saveEvery = 25;
28
29    thisDir = fileparts(mfilename('fullpath'));
30    datafile = fullfile(thisDir, 'fig1_data.mat');
```

This first block defines the function, clears the MATLAB workspace, and records the parameters from the paper. The period is $L = 6$, the perturbation size is $\epsilon = 10^{-4}$, and the two Fourier coefficients are $c_1 = 1/2$ and $c_{-1} = (0.3 - 0.4i)/2$. The code then sets $k_1 = 2\pi/L$, chooses $N_x = 384$, $T_{\max} = 60$, and $N_t = 30000$. The line `dt = Tmax/Nt` is the time step $\Delta t = 0.002$. The variable `thisDir` is used so that both the cached data file and the exported PNG are saved in the same folder as the MATLAB file.

```

1     if exist(datafile, 'file')
2
3         load(datafile, 'x', 'Tplot', 'Uplot', 'L', 'Tmax');
4
5     else
```



```

6      x = linspace(-L/2, L/2, Nx+1);
7      x(end) = [];
8
9
10     k = (2*pi/L)*[0:Nx/2-1, -Nx/2:-1];
11
12     u = 1 + eps0*( ...
13         c1 * exp( 1i*k1*x ) ...
14         + cm1 * exp(-1i*k1*x ) );
15
16     Ehalf = exp(-1i*k.^2*dt/2);
17
18     nSave = floor(Nt/saveEvery) + 1;
19
20     Uplot = zeros(nSave,Nx);
21     Tplot = zeros(nSave,1);
22
23     Uplot(1,:) = abs(u);
24     Tplot(1) = 0;

```

This block either loads previously computed data or starts a new simulation. If `fig1_data.mat` already exists, MATLAB loads `x`, `Tplot`, and `Uplot`; otherwise it constructs the periodic spatial grid and the FFT wave-number vector (3.12). The initial condition in line 47 is the discrete version of (3.8). Notice that the small factor ϵ appears as `eps0`, so the coefficients `c1` and `cm1` are the paper coefficients themselves, not already scaled coefficients. The array `Uplot` will store $|u|$, while `Tplot` stores the corresponding output times.

```

1      idx = 1;
2
3      for n = 1:Nt
4
5          % half linear step
6          uhat = fft(u);
7          uhat = Ehalf .* uhat;
8          u = ifft(uhat);
9
10         % nonlinear step
11         u = u .* exp(2i*abs(u).^2*dt);
12
13         % half linear step
14         uhat = fft(u);
15         uhat = Ehalf .* uhat;
16         u = ifft(uhat);
17
18         if mod(n,saveEvery)==0
19             idx = idx + 1;
20             Uplot(idx,:) = abs(u);
21             Tplot(idx) = n*dt;
22         end
23     end
24
25     save(datafile,'x','Tplot','Uplot','L','Tmax');
26
27 end

```

This is the numerical time stepping block. Lines 65–68 perform the first linear half-step: `fft(u)` computes the Fourier coefficients, multiplication by `Ehalf` applies the factor $\exp(-ik^2\Delta t/2)$, and `ifft` returns to physical space. Line 71 applies the nonlinear phase rotation (3.14). Lines 73–76 repeat the same Fourier half-step and complete one Strang step. The code stores the amplitude $|u|$ every `saveEvery = 25` time steps, so the plotted data are not too large but still show a smooth surface.

```

1      [Xgrid,Tgrid] = meshgrid(x,Tplot);
2
3      % Use software OpenGL in batch mode when available.
4      try
5          opengl software;
6      catch
7          end
8
9      % Changed back to v3-style size.
10     figure('Color','w','Position',[80 80 1400 460]);
11     set(gcf,'Renderer','opengl');
12     set(gcf,'InvertHardcopy','off');
13

```

```

14  %% =====
15  % Grayscale surface coloring
16  %% =====
17
18  grayLo = 0.44;
19  grayHi = 0.72;
20
21  Umin = 0.70;
22  Umax = 2.85;
23
24  Uclip = min(max(Uplot,Umin),Umax);
25  Uscaled = (Uclip - Umin)/(Umax - Umin);
26
27  Cgray = grayLo + (grayHi-grayLo)*Uscaled;
28
29  s = surf(Tgrid, Xgrid, Uplot, Cgray);
30
31  set(s, ...
32      'EdgeColor','none', ...
33      'FaceColor','interp', ...
34      'FaceLighting','gouraud', ...
35      'AmbientStrength',0.55, ...
36      'DiffuseStrength',0.85, ...
37      'SpecularStrength',0.18, ...
38      'SpecularExponent',25);
39
40  colormap(gray(256));
41  caxis([0 1]);

```

This block converts the saved numerical data into a surface plot. The horizontal axes are t and x , and the vertical axis is $|u(x, t)|$. The array Cgray is not new numerical data; it is only color information for the surface. It maps the amplitude into a moderate grayscale range so that the final image resembles the black-and-white figure in the paper instead of MATLAB's default colored surface plot.

```

1  xlim([-4 Tmax+4]);
2  ylim([-L/2 L/2+0.55]);
3  zlim([0 3.05]);
4
5  % Changed back to v3-style aspect ratio.
6  pbaspect([9.5 1.55 1.05]);
7
8  % Changed back to v3-style orientation.
9  view([33 18]);
10 camproj('orthographic');
11
12 % Use the v5 framing but remove the excess canvas after export.
13 camzoom(1.45);
14
15 axis off;
16 box off;
17
18 % Make the axes occupy almost the whole PNG.
19 set(gca,'Position',[-0.03 -0.03 1.08 1.08]);
20 set(gca,'LooseInset',[0 0 0 0]);
21
22 %% =====
23 % Lighting
24 %% =====
25
26 delete(findall(gca,'Type','light'));
27
28 light('Position',[-35 -25 35],'Style','infinite');
29 light('Position',[45 20 18],'Style','infinite');
30
31 lighting gouraud;
32 material dull;

```

This block controls the camera, aspect ratio, zoom, and lighting. These commands do not change the computed solution; they only change how the three-dimensional surface is viewed. The limits include the whole computed amplitude range, the aspect ratio makes the time direction long and thin as in the paper, and the orthographic camera avoids perspective distortion. The two light sources create the gray highlights and shadows that make the peaks easier to see.

```

1  hold on;
2

```

```

3  z0 = 0.07;
4  y0 = -L/2;
5
6  plot3([0 Tmax],[y0 y0],[z0 z0], ...
7        'k','LineWidth',1.15);
8
9  quiver3(0,y0,z0,-3.2,0,0,0, ...
10         'k','LineWidth',1.15,'MaxHeadSize',0.8);
11
12  quiver3(Tmax,y0,z0,0,L,0,0, ...
13         'k','LineWidth',1.15,'MaxHeadSize',0.8);
14
15  quiver3(Tmax,y0,z0,0,0,1.15,0, ...
16         'k','LineWidth',1.15,'MaxHeadSize',0.8);
17
18  text(-3.7,y0,z0,'t','FontSize',16);
19  text(Tmax,L/2+0.35,z0,'x','FontSize',16);
20
21  hold off;
22
23  %% =====
24  % Save and crop PNG
25  %% =====
26
27  drawnow;
28
29  set(gcf,'PaperPositionMode','auto');
30  rawFile = fullfile(thisDir,'paper_fig1_reproduction_bw_v6_raw.png');
31  outFile = fullfile(thisDir,'paper_fig1_reproduction_bw_v6.png');
32
33  print(gcf,rawFile,'-dpng','-r300');
34
35  rgb = imread(rawFile);
36  contentMask = any(rgb < 248,3);
37
38  rows = find(any(contentMask,2));
39  cols = find(any(contentMask,1));
40
41  if ~isempty(rows) && ~isempty(cols)
42      pad = 80;
43      r1 = max(min(rows)-pad,1);
44      r2 = min(max(rows)+pad,size(rgb,1));
45      c1 = max(min(cols)-pad,1);
46      c2 = min(max(cols)+pad,size(rgb,2));
47      rgb = rgb(r1:r2,c1:c2,:);
48  end
49
50  imwrite(rgb,outFile);
51
52  if exist(rawFile,'file')
53      delete(rawFile);
54  end

```

The last block draws simple black coordinate arrows and exports the PNG file. The arrows are added with `plot3`, `quiver3`, and `text`; they are visual annotations, not part of the numerical method. The v6 script first writes a temporary raw PNG, reads it back with `imread`, detects the non-white part of the image, and crops away the large empty white margins. It then writes the final v6 PNG. This changes only the exported canvas, not the computed solution.

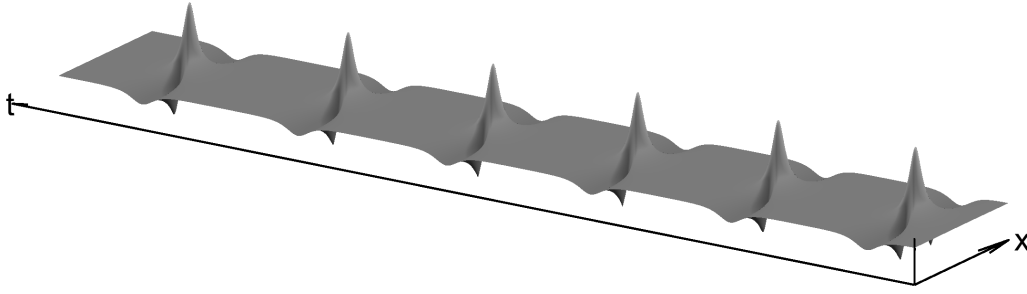


FIGURE 1. Cropped v6 MATLAB export of the paper-style recurrence.

Figure 1 shows the expected anomalous-wave recurrence. For small t , the surface is close to the background $|u| = 1$. The one unstable Fourier mode then grows, a sharp nonlinear peak forms, and the solution returns close to the background before the next growth stage. The important point is that the large peak is not inserted by hand: it is generated by repeatedly applying the three Strang-splitting operations described above.

Supplementary views and stability checks. The earlier colored view is kept because it gives a complementary way to read the recurrence times: the nearly uniform background corresponds to $|u| \approx 1$, while the localized high-amplitude regions correspond to the nonlinear peaks in the paper-style surface. To check that the large peaks are not merely plotting artifacts, we also ran the same type of split-step Fourier simulation in two stable regimes. In the first stable test, the code keeps a fine grid and small time step, but changes the length to $L = 2$. Then

$$(3.20) \quad k_1 = \frac{2\pi}{L} = \pi > 2.$$

Thus the first Fourier mode is outside the modulationally unstable band. With a small perturbation and conservative numerical parameters, the solution remains close to the unit background. The under-resolved test uses a coarser grid $N_x = 256$, a much larger final time $T_{\max} = 250$, and only $N_t = 20000$ time steps; its irregular spikes should therefore be interpreted as numerical drift or accumulated discretization error, not as physical modulation instability. The three checks are collected compactly in Figure 2.

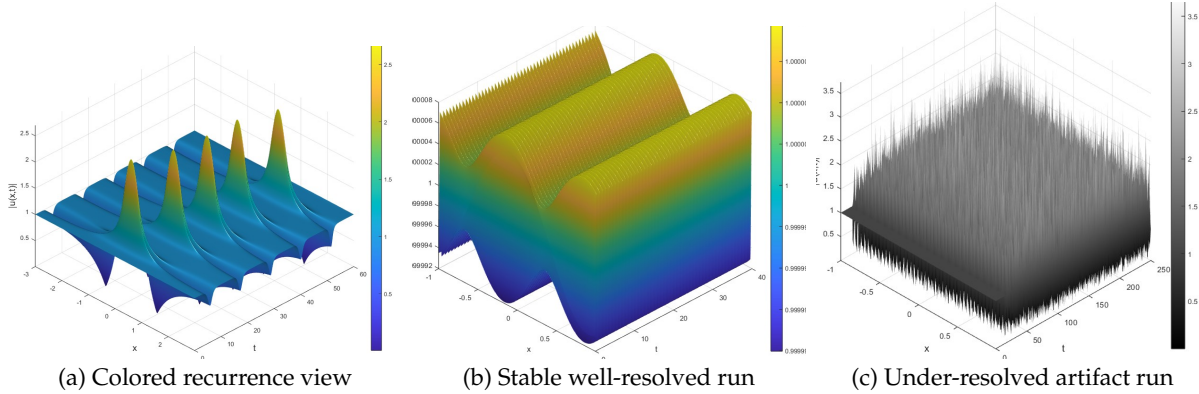


FIGURE 2. Supplementary views and stability checks. Panel (a) gives a colored view of the one-mode recurrence, panel (b) shows a stable well-resolved run, and panel (c) shows that coarse long-time simulations can create spurious peaks even in a stable regime.

This compact comparison separates the genuine one-mode recurrence in Figure 1 from artifacts caused by insufficient numerical resolution.

Note 1: Linear Stage in the Grinevich–Santini Recurrence Paper

2026-05-27

In this note we organize the part of the calculation in P. G. Grinevich and P. M. Santini, *The finite gap method and the analytic description of the exact rogue wave recurrence in the periodic NLS Cauchy problem*, Nonlinearity **31** (2018), 5258–5308, which leads to formulas (15) and (16) of the paper. The purpose here is only to understand the first linear stage of modulation instability. The nonlinear Akhmediev-breather formula (18)–(19) will be used later, but we do not derive the full finite-gap in this note.

Setting and unstable Fourier modes. The equation is the focusing nonlinear Schrödinger equation

$$(N1.1) \quad iu_t + u_{xx} + 2|u|^2u = 0, \quad u = u(x, t) \in \mathbb{C},$$

on the periodic interval of length L . The background solution is

$$(N1.2) \quad u_0(x, t) = e^{2it}.$$

The initial condition is a small perturbation of this background:

$$(N1.3) \quad u(x, 0) = 1 + \epsilon v(x), \quad 0 < \epsilon \ll 1,$$

where

$$(N1.4) \quad v(x) = \sum_{j \geq 1} \left(c_j e^{ik_j x} + c_{-j} e^{-ik_j x} \right), \quad k_j = \frac{2\pi j}{L}.$$

The j -th Fourier mode is the pair of waves $e^{ik_j x}$ and $e^{-ik_j x}$. These two waves must be treated together, because the linearized NLS equation contains the complex conjugate of the perturbation.

For one unstable mode n , the paper assumes

$$(N1.5) \quad v(x) = c_n e^{ik_n x} + c_{-n} e^{-ik_n x}.$$

The mode is unstable precisely when

$$(N1.6) \quad 0 < k_n < 2.$$

It is useful to introduce the angle

$$(N1.7) \quad \phi_n = \arccos \frac{k_n}{2} = \arccos \frac{\pi n}{L}, \quad 0 < \phi_n < \frac{\pi}{2}.$$

Then

$$(N1.8) \quad k_n = 2 \cos \phi_n.$$

Linearization about the background. Write the solution near the background as

$$(N1.9) \quad u(x, t) = e^{2it} (1 + w(x, t)), \quad |w| \ll 1.$$

We compute the terms in (N1.1). First,

$$(N1.10) \quad u_t = 2ie^{2it}(1 + w) + e^{2it}w_t, \quad u_{xx} = e^{2it}w_{xx}.$$

Next,

$$(N1.11) \quad \begin{aligned} |u|^2 u &= e^{2it} |1 + w|^2 (1 + w) \\ &= e^{2it} (1 + 2w + \bar{w}) + O(|w|^2). \end{aligned}$$

Substituting these expressions into (N1.1) and keeping only the terms linear in w , the constant background terms cancel and we obtain

$$(N1.12) \quad iw_t + w_{xx} + 2(w + \bar{w}) = 0.$$

Growth rate. Because of the term $\bar{w}$, a single wave e^{ikx} is coupled to e^{-ikx} . Therefore, for a fixed wave number k , we set

$$(N1.13) \quad w(x, t) = ae^{ikx + \sigma t} + be^{-ikx + \sigma t}.$$

Here σ is the growth or decay exponent. Substituting (N1.13) into (N1.12) and comparing the coefficients of $e^{ikx+\sigma t}$ and $e^{-ikx+\sigma t}$, we get

$$(N1.14) \quad \begin{pmatrix} 2 - k^2 + i\sigma & 2 \\ 2 & 2 - k^2 - i\sigma \end{pmatrix} \begin{pmatrix} a \\ \bar{b} \end{pmatrix} = 0.$$

For a nonzero solution, the determinant must vanish:

$$(N1.15) \quad (2 - k^2 + i\sigma)(2 - k^2 - i\sigma) - 4 = 0.$$

Hence

$$(N1.16) \quad \sigma^2 = 4 - (2 - k^2)^2 = k^2(4 - k^2).$$

Thus the unstable growth rate is

$$(N1.17) \quad \sigma(k) = k\sqrt{4 - k^2}, \quad 0 < k < 2.$$

For the n -th unstable mode,

$$(N1.18) \quad \sigma_n = k_n \sqrt{4 - k_n^2} = 2 \sin(2\phi_n),$$

where we used $k_n = 2 \cos \phi_n$.

Growing and decaying coordinates. The initial Fourier coefficients c_n, c_{-n} measure how much of the positive and negative wave numbers is present. They are not themselves the growing and decaying amplitudes. The linearized system diagonalizes into one direction proportional to $e^{\sigma_n t}$ and one direction proportional to $e^{-\sigma_n t}$.

We now compute these two directions explicitly. First consider a pure growing solution of (N1.12):

$$(N1.19) \quad w_g(x, t) = e^{\sigma_n t} \left(a e^{ik_n x} + b e^{-ik_n x} \right).$$

The coefficient of $e^{ik_n x + \sigma_n t}$ gives

$$(N1.20) \quad (2 - k_n^2 + i\sigma_n) a + 2\bar{b} = 0.$$

Therefore

$$(N1.21) \quad \bar{b} = -\frac{2 - k_n^2 + i\sigma_n}{2} a.$$

Using

$$(N1.22) \quad k_n = 2 \cos \phi_n, \quad \sigma_n = 2 \sin(2\phi_n),$$

we have

$$(N1.23) \quad 2 - k_n^2 = -2 \cos(2\phi_n).$$

Thus

$$(N1.24) \quad -\frac{2 - k_n^2 + i\sigma_n}{2} = \cos(2\phi_n) - i \sin(2\phi_n) = e^{-2i\phi_n}.$$

Hence

$$(N1.25) \quad \bar{b} = e^{-2i\phi_n} a, \quad b = e^{2i\phi_n} \bar{a}.$$

This means that the two Fourier coefficients in a pure growing solution cannot be chosen independently.

If we call the coefficient of $e^{-ik_n x}$ in the growing part A , then the corresponding coefficient of $e^{ik_n x}$ is $e^{2i\phi_n} \bar{A}$. Therefore the growing part may be written as

$$(N1.26) \quad w_g(x, t) = e^{\sigma_n t} \left(A e^{-ik_n x} + e^{2i\phi_n} \bar{A} e^{ik_n x} \right).$$

Similarly, a pure decaying solution has the form

$$(N1.27) \quad w_d(x, t) = e^{-\sigma_n t} \left(c e^{ik_n x} + d e^{-ik_n x} \right).$$

Now the coefficient of $e^{ik_n x - \sigma_n t}$ gives

$$(N1.28) \quad (2 - k_n^2 - i\sigma_n) c + 2\bar{d} = 0.$$

Using the same trigonometric identities,

$$(N1.29) \quad \bar{d} = -\frac{2 - k_n^2 - i\sigma_n}{2} c = e^{2i\phi_n} c, \quad d = e^{-2i\phi_n} \bar{c}.$$

If we call the coefficient of $e^{ik_n x}$ in the decaying part D , then

$$(N1.30) \quad w_d(x, t) = e^{-\sigma_n t} \left(e^{-2i\phi_n} \bar{D} e^{-ik_n x} + D e^{ik_n x} \right).$$

The general linear solution in this unstable mode is therefore

$$(N1.31) \quad w(x, t) = e^{\sigma_n t} \left(A e^{-ik_n x} + e^{2i\phi_n} \bar{A} e^{ik_n x} \right) + e^{-\sigma_n t} \left(e^{-2i\phi_n} \bar{D} e^{-ik_n x} + D e^{ik_n x} \right).$$

At $t = 0$, this must match the initial perturbation

$$(N1.32) \quad w(x, 0) = \epsilon \left(c_n e^{ik_n x} + c_{-n} e^{-ik_n x} \right).$$

Comparing the coefficients of $e^{-ik_n x}$ and $e^{ik_n x}$ gives

$$(N1.33) \quad A + e^{-2i\phi_n} \bar{D} = \epsilon c_{-n},$$

$$(N1.34) \quad e^{2i\phi_n} \bar{A} + D = \epsilon c_n.$$

Now solve these two equations. Taking the complex conjugate of (N1.34) gives

$$(N1.35) \quad e^{-2i\phi_n} A + \bar{D} = \epsilon \bar{c}_n.$$

Subtracting $e^{-2i\phi_n}$ times (N1.35) from (N1.33), we obtain

$$(N1.36) \quad (1 - e^{-4i\phi_n}) A = \epsilon \left(c_{-n} - e^{-2i\phi_n} \bar{c}_n \right).$$

Since

$$(N1.37) \quad 1 - e^{-4i\phi_n} = e^{-2i\phi_n} (e^{2i\phi_n} - e^{-2i\phi_n}) = i\sigma_n e^{-2i\phi_n},$$

and

$$(N1.38) \quad c_{-n} - e^{-2i\phi_n} \bar{c}_n = -e^{-2i\phi_n} (\bar{c}_n - e^{2i\phi_n} c_{-n}),$$

we get

$$(N1.39) \quad A = \frac{i}{\sigma_n} \epsilon \left(\bar{c}_n - e^{2i\phi_n} c_{-n} \right).$$

This is why the paper defines

$$(N1.40) \quad \alpha_n = \epsilon \left(\bar{c}_n - e^{2i\phi_n} c_{-n} \right).$$

Thus

$$(N1.41) \quad A = \frac{i\alpha_n}{\sigma_n}.$$

The decaying coefficient is found in the same way. Taking the complex conjugate of (N1.33) gives

$$(N1.42) \quad \bar{A} + e^{2i\phi_n} D = \epsilon \bar{c}_{-n}.$$

Subtracting $e^{-2i\phi_n}$ times (N1.34) from (N1.42), we obtain

$$(N1.43) \quad (e^{2i\phi_n} - e^{-2i\phi_n}) D = \epsilon \left(\bar{c}_{-n} - e^{-2i\phi_n} c_n \right).$$

Since

$$e^{2i\phi_n} - e^{-2i\phi_n} = i\sigma_n,$$

we get

$$(N1.44) \quad D = \frac{1}{i\sigma_n} \epsilon \left(\overline{c_{-n}} - e^{-2i\phi_n} c_n \right).$$

This is why the paper defines

$$(N1.45) \quad \beta_n = \epsilon \left(\overline{c_{-n}} - e^{-2i\phi_n} c_n \right).$$

Thus

$$(N1.46) \quad D = -\frac{i\beta_n}{\sigma_n}.$$

Combining (N1.40) and (N1.45), the paper's coefficients are

$$(N1.47) \quad \alpha_n = \epsilon \left(\overline{c_n} - e^{2i\phi_n} c_{-n} \right), \quad \beta_n = \epsilon \left(\overline{c_{-n}} - e^{-2i\phi_n} c_n \right).$$

They are $O(\epsilon)$. The coefficient α_n is the growing coordinate of the initial perturbation, while β_n is the decaying coordinate.

Substituting $A = i\alpha_n/\sigma_n$ and $D = -i\beta_n/\sigma_n$ into (N1.31), we obtain

$$(N1.48) \quad \begin{aligned} w(x, t) = & \frac{i}{\sigma_n} \left(\alpha_n e^{-ik_n x} - e^{2i\phi_n} \overline{\alpha_n} e^{ik_n x} \right) e^{\sigma_n t} \\ & - \frac{i}{\sigma_n} \left(\beta_n e^{ik_n x} - e^{-2i\phi_n} \overline{\beta_n} e^{-ik_n x} \right) e^{-\sigma_n t}. \end{aligned}$$

The first line is the growing α_n -wave, and the second line is the decaying β_n -wave.

The growing wave as a cosine. Write

$$(N1.49) \quad \alpha_n = |\alpha_n| e^{i\theta_\alpha}, \quad \theta_\alpha = \arg(\alpha_n).$$

The growing part of (N1.48) is

$$(N1.50) \quad w_{\text{grow}} = \frac{i}{\sigma_n} \left(\alpha_n e^{-ik_n x} - e^{2i\phi_n} \overline{\alpha_n} e^{ik_n x} \right) e^{\sigma_n t}.$$

We want to write it as a shifted cosine. Define

$$(N1.51) \quad X_n^+ = \frac{\arg(\alpha_n) - \phi_n + \pi/2}{k_n}.$$

Then

$$(N1.52) \quad k_n X_n^+ = \theta_\alpha - \phi_n + \frac{\pi}{2},$$

and

$$(N1.53) \quad \begin{aligned} \frac{2|\alpha_n|}{\sigma_n} e^{\sigma_n t + i\phi_n} \cos(k_n(x - X_n^+)) &= \frac{|\alpha_n|}{\sigma_n} e^{\sigma_n t + i\phi_n} \left[e^{ik_n(x - X_n^+)} + e^{-ik_n(x - X_n^+)} \right] \\ &= \frac{i}{\sigma_n} \left(\alpha_n e^{-ik_n x} - e^{2i\phi_n} \overline{\alpha_n} e^{ik_n x} \right) e^{\sigma_n t}. \end{aligned}$$

Therefore

$$(N1.54) \quad w_{\text{grow}} = \frac{2|\alpha_n|}{\sigma_n} \cos(k_n(x - X_n^+)) e^{\sigma_n t + i\phi_n}.$$

The point X_n^+ is the position of the maximum of the growing sinusoidal wave.

The decaying wave as a cosine. Similarly, write

$$(N1.55) \quad \beta_n = |\beta_n| e^{i\theta_\beta}, \quad \theta_\beta = \arg(\beta_n).$$

Define

$$(N1.56) \quad X_n^- = \frac{-\arg(\beta_n) - \phi_n + \pi/2}{k_n}.$$

Then the decaying part of (N1.48) becomes

$$(N1.57) \quad w_{\text{decay}} = \frac{2|\beta_n|}{\sigma_n} \cos(k_n(x - X_n^-)) e^{-\sigma_n t - i\phi_n}.$$

The point X_n^- is the position of the maximum of the decaying sinusoidal wave.

Equations (N1.51) and (N1.56) are precisely formula (16) in the paper:

$$(N1.58) \quad X_n^+ = \frac{\arg(\alpha_n) - \phi_n + \pi/2}{k_n}, \quad X_n^- = \frac{-\arg(\beta_n) - \phi_n + \pi/2}{k_n}.$$

Formula (15). Combining (N1.54) and (N1.57), we obtain

$$(N1.59) \quad w(x, t) = \frac{2}{\sigma_n} \left[|\alpha_n| \cos(k_n(x - X_n^+)) e^{\sigma_n t + i\phi_n} + |\beta_n| \cos(k_n(x - X_n^-)) e^{-\sigma_n t - i\phi_n} \right].$$

Since $u = e^{2it}(1 + w)$, this gives

$$(N1.60) \quad u(x, t) = e^{2it} \left\{ 1 + \frac{2}{\sigma_n} \left[|\alpha_n| \cos(k_n(x - X_n^+)) e^{\sigma_n t + i\phi_n} + |\beta_n| \cos(k_n(x - X_n^-)) e^{-\sigma_n t - i\phi_n} \right] \right\} + O(\epsilon^2 |\log \epsilon|).$$

This is formula (15) of the paper. The interpretation is simple: the unstable part of the initial datum splits into a growing α_n -wave and a decaying β_n -wave.

The remainder in (N1.60) should not be confused with the ordinary error one would get by only linearizing the PDE. The linearized calculation explains the main $O(\epsilon)$ terms. The stated $O(\epsilon^2 |\log \epsilon|)$ remainder comes from the finite-gap construction: the small gaps have size $O(\epsilon)$, and the asymptotic expansion of the Riemann matrix and theta functions contains logarithmic error estimates of the form $O(\epsilon^2 |\log \epsilon|)$. In the early time region $0 \leq t \leq O(1)$, the terms $|\alpha_n| e^{\sigma_n t}$ and $|\beta_n| e^{-\sigma_n t}$ are still $O(\epsilon)$, so this linear-stage formula is valid.

Bridge to the first nonlinear stage. Formula (N1.60) is not intended to describe the moment when the growing wave has become $O(1)$. That happens on the logarithmic time scale

$$(N1.61) \quad t = O\left(\frac{|\log \epsilon|}{\sigma_n}\right).$$

At that scale the growing α_n -wave triggers the first nonlinear stage. We now show how the appearance time is obtained by matching the growing wave in (N1.60) with the left tail of the Akhmediev breather.

For this matching calculation, abbreviate

$$\sigma = \sigma_n, \quad k = k_n, \quad \phi = \phi_n, \quad X = X_n^+.$$

The Akhmediev breather used for the first nonlinear stage has the form

$$(N1.62) \quad A(x, t; \phi, X, T, 2\phi) = e^{2it + 2i\phi} \frac{\cosh(\sigma(t - T) + 2i\phi) + \sin \phi \cos(k(x - X))}{\cosh(\sigma(t - T)) - \sin \phi \cos(k(x - X))}.$$

Here T is still unknown. The left tail means that the current time is well before the breather center time T , on the time scale $1/\sigma$:

$$(N1.63) \quad \sigma(t - T) \ll -1, \quad \text{equivalently} \quad T - t \gg \frac{1}{\sigma}.$$

This condition does not require $t < 0$. It only means that t is far to the left of the peak time T .

Set

$$(N1.64) \quad y = \sigma(t - T), \quad C = \cos(k(x - X)), \quad s = \sin \phi, \quad q = e^y.$$

In the left tail, $q \ll 1$. Formula (N1.62) becomes

$$(N1.65) \quad A = e^{2it+2i\phi} \frac{\cosh(y+2i\phi) + sC}{\cosh y - sC}.$$

We expand the denominator first. Since

$$(N1.66) \quad \cosh y = \frac{e^y + e^{-y}}{2} = \frac{q + q^{-1}}{2} = \frac{1}{2q} (1 + q^2),$$

we have

$$(N1.67) \quad \cosh y - sC = \frac{1}{2q} (1 - 2sCq + q^2).$$

For the numerator,

$$(N1.68) \quad \begin{aligned} \cosh(y+2i\phi) &= \frac{e^{y+2i\phi} + e^{-y-2i\phi}}{2} \\ &= \frac{1}{2q} (e^{-2i\phi} + q^2 e^{2i\phi}). \end{aligned}$$

Therefore

$$(N1.69) \quad \cosh(y+2i\phi) + sC = \frac{1}{2q} (e^{-2i\phi} + 2sCq + q^2 e^{2i\phi}).$$

Dividing (N1.69) by (N1.67), the common factor $1/(2q)$ cancels:

$$(N1.70) \quad A = e^{2it+2i\phi} \frac{e^{-2i\phi} + 2sCq + q^2 e^{2i\phi}}{1 - 2sCq + q^2}.$$

Keeping only terms up to first order in q , we use

$$\frac{1}{1 - 2sCq + O(q^2)} = 1 + 2sCq + O(q^2).$$

Thus

$$(N1.71) \quad A = e^{2it} \left[1 + 2sC (1 + e^{2i\phi}) q + O(q^2) \right].$$

The coefficient simplifies because

$$(N1.72) \quad 1 + e^{2i\phi} = 2 \cos \phi e^{i\phi}, \quad 2s (1 + e^{2i\phi}) = 2 \sin(2\phi) e^{i\phi} = \sigma e^{i\phi}.$$

Since $q = e^{\sigma(t-T)}$, the left-tail expansion is

$$(N1.73) \quad A(x, t; \phi, X, T, 2\phi) = e^{2it} \left[1 + \sigma e^{\sigma(t-T)+i\phi} \cos(k(x-X)) + O(e^{2\sigma(t-T)}) \right].$$

So, far before its center time, the Akhmediev breather looks like a small exponentially growing linear wave on top of the background.

Now set $\phi = \phi_n$, $k = k_n$, $X = X_n^+$, and $\sigma = \sigma_n$. The growing part of the linear formula (N1.60) is

$$(N1.74) \quad u_{\text{lin}}(x, t) \sim e^{2it} \left[1 + \frac{2|\alpha_n|}{\sigma_n} e^{\sigma_n t + i\phi_n} \cos(k_n(x - X_n^+)) \right].$$

The left tail (N1.73) gives

$$(N1.75) \quad A(x, t; \phi_n, X_n^+, T, 2\phi_n) \sim e^{2it} \left[1 + \sigma_n e^{\sigma_n(t-T)+i\phi_n} \cos(k_n(x - X_n^+)) \right].$$

Rewrite the coefficient in (N1.75) as

$$(N1.76) \quad \sigma_n e^{\sigma_n(t-T)+i\phi_n} = \sigma_n e^{-\sigma_n T} e^{\sigma_n t + i\phi_n}.$$

Therefore the breather tail has growing-wave coefficient $\sigma_n e^{-\sigma_n T}$, while the linear solution has coefficient $2|\alpha_n|/\sigma_n$. Matching these coefficients gives

$$(N1.77) \quad \frac{2|\alpha_n|}{\sigma_n} = \sigma_n e^{-\sigma_n T}.$$

Solving this equation,

$$(N1.78) \quad e^{-\sigma_n T} = \frac{2|\alpha_n|}{\sigma_n^2}, \quad -\sigma_n T = \log \left(\frac{2|\alpha_n|}{\sigma_n^2} \right).$$

Hence

$$(N1.79) \quad T_n(|\alpha_n|) = \frac{1}{\sigma_n} \log \left(\frac{\sigma_n^2}{2|\alpha_n|} \right),$$

which is formula (17) in the paper. The spatial center is X_n^+ . Thus the modulus $|\alpha_n|$ determines the first appearance time, while $\arg(\alpha_n)$ determines the position of the first peak.

Note 2: Periodic Spectral Theory and the Monodromy Matrix

2026-06-03

This note begins the part of Section 3 of Grinevich–Santini concerned with the periodic spectral problem. The purpose is to make precise the objects that enter the finite-gap description of the periodic NLS Cauchy problem: the fundamental matrix, the monodromy matrix, Bloch multipliers, the main spectrum, and the auxiliary spectrum.

The fixed-time spatial spectral problem. The focusing NLS equation

$$(N2.1) \quad iu_t + u_{xx} + 2|u|^2 u = 0$$

is represented by the compatibility of the Lax pair

$$(N2.2) \quad \Psi_x(\lambda, x, t) = U(\lambda, x, t)\Psi(\lambda, x, t), \quad \Psi_t(\lambda, x, t) = V(\lambda, x, t)\Psi(\lambda, x, t),$$

where the spatial matrix is

$$(N2.3) \quad U(\lambda, x, t) = \begin{pmatrix} -i\lambda & iu(x, t) \\ i\overline{u(x, t)} & i\lambda \end{pmatrix}.$$

For the construction of the periodic spectrum, first fix a time $t = t_0$. Then λ is a spectral parameter and (N2.2) becomes an ordinary differential equation in the spatial variable x :

$$(N2.4) \quad \Psi_x(\lambda, x, t_0) = U(\lambda, x, t_0)\Psi(\lambda, x, t_0).$$

If the potential is L -periodic,

$$(N2.5) \quad u(x + L, t_0) = u(x, t_0),$$

then $U(\lambda, x + L, t_0) = U(\lambda, x, t_0)$. Thus, after fixing λ and t_0 , the spatial problem has the form

$$(N2.6) \quad \Psi_x = A(x)\Psi, \quad A(x + L) = A(x),$$

with $A(x) = U(\lambda, x, t_0)$.

Equivalently, the spatial equation can be written as the spectral problem

$$(N2.7) \quad L\Psi = \lambda\Psi, \quad L = \begin{pmatrix} i\partial_x & u \\ -\overline{u} & -i\partial_x \end{pmatrix}.$$

This operator is not self-adjoint in the focusing case. Consequently its spectrum is not restricted to the real axis, and the periodic theory is built from Floquet theory and analytic matrix functions rather than from the self-adjoint spectral theorem.

Fundamental matrix and propagation. For any reference point y , define the fundamental matrix by the initial-value problem

$$(N2.8) \quad \frac{\partial}{\partial x} \Phi(\lambda; x, y, t_0) = U(\lambda, x, t_0)\Phi(\lambda; x, y, t_0), \quad \Phi(\lambda; y, y, t_0) = I.$$

When λ and t_0 are fixed, we write simply $\Phi(x, y)$. The matrix $\Phi(x, y)$ propagates initial data from y to x : if $\Psi(y) = v$, then

$$(N2.9) \quad \Psi(x) = \Phi(x, y)v.$$

The two columns of $\Phi(x, y)$ are the two solutions whose initial data at $x = y$ are the standard basis vectors.

The fundamental matrix satisfies the composition law

$$(N2.10) \quad \Phi(x, z) = \Phi(x, y)\Phi(y, z).$$

Indeed, for fixed y and z , both $\Phi(x, z)$ and $\Phi(x, y)\Phi(y, z)$ solve the same matrix ODE in x , and at $x = y$ both are equal to $\Phi(y, z)$. Uniqueness of the initial-value problem gives (N2.10).

Taking $z = x$ in (N2.10) gives the inverse evolution identity

$$(N2.11) \quad \Phi(x, y)^{-1} = \Phi(y, x).$$

The periodicity of U gives another basic identity:

$$(N2.12) \quad \Phi(x + L, y + L) = \Phi(x, y).$$

To see this, define $F(x) = \Phi(x + L, y + L)$. Then

$$F_x(x) = U(\lambda, x + L, t_0)F(x) = U(\lambda, x, t_0)F(x),$$

and $F(y) = I$. This is the same initial-value problem as the one for $\Phi(x, y)$, hence (N2.12) follows.

The monodromy matrix. Fix a base point x_0 . The monodromy matrix is the fundamental matrix over one spatial period:

$$(N2.13) \quad T(\lambda; x_0, t_0) = \Phi(\lambda; x_0 + L, x_0, t_0).$$

Equivalently, if $\Psi(x_0) = v$, then

$$(N2.14) \quad \Psi(x_0 + L) = T(\lambda; x_0, t_0)v.$$

Thus T records the total effect of the periodic spatial equation after going once around the period.

The identity

$$(N2.15) \quad \Phi(x + L, x_0) = \Phi(x, x_0)T$$

is a direct consequence of the composition law and periodicity. Namely,

$$\Phi(x + L, x_0) = \Phi(x + L, x_0 + L)\Phi(x_0 + L, x_0),$$

and the two factors on the right are, respectively, $\Phi(x, x_0)$ and T .

Changing the base point conjugates the monodromy matrix. If x_1 is another base point, then

$$(N2.16) \quad T(\lambda; x_1, t_0) = \Phi(x_1, x_0)T(\lambda; x_0, t_0)\Phi(x_0, x_1).$$

Here is the computation. Starting from the definition at the new base point,

$$T(\lambda; x_1, t_0) = \Phi(x_1 + L, x_1).$$

Split the propagation from x_1 to $x_1 + L$ through the intermediate points x_0 and $x_0 + L$:

$$\Phi(x_1 + L, x_1) = \Phi(x_1 + L, x_0 + L)\Phi(x_0 + L, x_0)\Phi(x_0, x_1).$$

By periodicity, $\Phi(x_1 + L, x_0 + L) = \Phi(x_1, x_0)$, and by definition $\Phi(x_0 + L, x_0) = T(\lambda; x_0, t_0)$. This gives (N2.16). Since $\Phi(x_0, x_1) = \Phi(x_1, x_0)^{-1}$, the two monodromy matrices are conjugate. Therefore the eigenvalues of T , and hence the spectral curve below, do not depend on the chosen base point.

Bloch solutions and Floquet multipliers. Let $v \neq 0$ be an eigenvector of the monodromy matrix:

$$(N2.17) \quad Tv = \kappa v.$$

Define

$$(N2.18) \quad \Psi(x) = \Phi(x, x_0)v.$$

Using (N2.15), we get

$$(N2.19) \quad \begin{aligned} \Psi(x + L) &= \Phi(x + L, x_0)v \\ &= \Phi(x, x_0)Tv = \kappa\Phi(x, x_0)v = \kappa\Psi(x). \end{aligned}$$

Thus an eigenvector of the monodromy matrix at one base point produces a solution satisfying

$$(N2.20) \quad \Psi(x + L) = \kappa\Psi(x).$$

The number κ is a Floquet multiplier, or Bloch multiplier, and Ψ is a Bloch solution. Repeating the same relation gives

$$(N2.21) \quad \Psi(x + mL) = \kappa^m \Psi(x), \quad m \in \mathbb{Z}.$$

Hence $|\kappa| = 1$ corresponds to bounded oscillation from period to period, while $|\kappa| \neq 1$ corresponds to exponential growth in one spatial direction and decay in the other.

It is convenient to write

$$(N2.22) \quad \kappa = e^{iLp},$$

where p is the quasimomentum. If $\Psi(x + L) = e^{iLp}\Psi(x)$, then

$$(N2.23) \quad q(x) = e^{-ipx}\Psi(x)$$

satisfies $q(x + L) = q(x)$. Indeed,

$$q(x + L) = e^{-ip(x+L)} e^{iLp}\Psi(x) = e^{-ipx}\Psi(x) = q(x).$$

Thus a Bloch solution has the Floquet form

$$(N2.24) \quad \Psi(x) = e^{ipx}q(x), \quad q(x + L) = q(x).$$

Moreover,

$$(N2.25) \quad |\kappa| = |e^{iLp}| = e^{-L\operatorname{Im} p},$$

so $|\kappa| = 1$ is equivalent to

$$(N2.26) \quad \operatorname{Im} p = 0.$$

Determinant and reciprocal multipliers. The spatial matrix (N2.3) is traceless:

$$(N2.27) \quad \operatorname{tr} U(\lambda, x, t_0) = 0.$$

By Liouville's formula for matrix ODEs,

$$(N2.28) \quad \frac{\partial}{\partial x} \det \Phi(x, y) = \operatorname{tr} U(\lambda, x, t_0) \det \Phi(x, y).$$

Therefore $\det \Phi(x, y)$ is constant in x . Since $\Phi(y, y) = I$, we obtain

$$(N2.29) \quad \det \Phi(x, y) = 1.$$

In particular,

$$(N2.30) \quad \det T(\lambda; x_0, t_0) = 1.$$

If the two Floquet multipliers are denoted by κ_+ and κ_- , then

$$(N2.31) \quad \kappa_+(\lambda)\kappa_-(\lambda) = 1.$$

Thus the two sheets of the spectral curve correspond to reciprocal Floquet multipliers.

Floquet discriminant and the spectral curve. Define the Floquet discriminant by

$$(N2.32) \quad \Delta(\lambda) = \operatorname{tr} T(\lambda; x_0, t_0).$$

The Floquet multipliers are the eigenvalues of T , so they are the roots of the characteristic polynomial. For a 2×2 matrix,

$$\det(\kappa I - T) = \kappa^2 - \operatorname{tr} T \kappa + \det T = \kappa^2 - \Delta(\lambda)\kappa + 1,$$

where we used $\operatorname{tr} T = \Delta(\lambda)$ and $\det T = 1$. Thus the characteristic equation is

$$(N2.33) \quad \kappa^2 - \Delta(\lambda)\kappa + 1 = 0.$$

Therefore

$$(N2.34) \quad \kappa_{\pm}(\lambda) = \frac{\Delta(\lambda) \pm \sqrt{\Delta(\lambda)^2 - 4}}{2}.$$

The square root becomes single-valued after passing to the two-sheeted spectral curve

$$(N2.35) \quad \nu^2 = \Delta(\lambda)^2 - 4.$$

A point on the spectral curve is written as $\gamma = (\lambda, \nu)$, and the corresponding multiplier is

$$(N2.36) \quad \kappa(\gamma) = \frac{\Delta(\lambda) + \nu}{2}.$$

The two points above the same value of λ give the two reciprocal multipliers.

Periodic and antiperiodic points. A periodic eigenfunction satisfies

$$(N2.37) \quad \Psi(x + L) = \Psi(x),$$

so its multiplier is $\kappa = 1$. An antiperiodic eigenfunction satisfies

$$(N2.38) \quad \Psi(x + L) = -\Psi(x),$$

so its multiplier is $\kappa = -1$. Since $\kappa + \kappa^{-1} = \Delta(\lambda)$, the periodic and antiperiodic points are characterized by

$$(N2.39) \quad \Delta(\lambda) = 2 \quad \text{and} \quad \Delta(\lambda) = -2,$$

or, equivalently,

$$(N2.40) \quad \Delta(\lambda)^2 - 4 = 0.$$

These are the points where the two Floquet multipliers coincide: $\kappa_+ = \kappa_- = \pm 1$. If the zero of $\Delta^2 - 4$ is simple, the point is a branch point of the spectral curve. Even-order zeros may instead be double or resonant points.

Main spectrum. The main spectrum is the spectrum of the spatial operator on the line. In Floquet terms, it is the projection to the λ -plane of the points on the spectral curve whose Bloch multiplier has unit modulus:

$$(N2.41) \quad \lambda \in \sigma_{\text{main}} \iff |\kappa(\gamma)| = 1 \quad \text{for some } \gamma \text{ above } \lambda.$$

Equivalently, using $\kappa = e^{iLp}$,

$$(N2.42) \quad \text{Im } p(\gamma) = 0.$$

The corresponding Bloch functions are generalized eigenfunctions; they need not be in $L^2(\mathbb{R})$, just as plane waves for a constant coefficient operator are not square integrable on the whole line. Since the focusing NLS spectral operator is not self-adjoint, the main spectrum may contain complex arcs, not only real intervals.

Auxiliary spectrum and divisor. The main spectrum determines the spectral curve, but not the point of the solution inside the isospectral family. For this additional data one uses the auxiliary spectrum, also called the divisor. In the convention used in the paper, an auxiliary point is a point $\gamma \in \Gamma$ such that the first component of the Bloch eigenfunction vanishes at the base point:

$$(N2.43) \quad \Psi_1(\gamma, x_0, t) = 0.$$

Because the same eigenfunction satisfies the Bloch relation, this also implies

$$(N2.44) \quad \Psi_1(\gamma, x_0 + L, t) = \kappa(\gamma) \Psi_1(\gamma, x_0, t) = 0.$$

Thus the auxiliary spectrum is equivalently a Dirichlet-type spectrum on $[x_0, x_0 + L]$:

$$(N2.45) \quad \Psi_1(\lambda, x_0, t) = 0, \quad \Psi_1(\lambda, x_0 + L, t) = 0.$$

In terms of the monodromy matrix, write

$$(N2.46) \quad T(\lambda) = \begin{pmatrix} T_{11}(\lambda) & T_{12}(\lambda) \\ T_{21}(\lambda) & T_{22}(\lambda) \end{pmatrix}.$$

To see why the off-diagonal entry appears, choose the solution whose initial value at x_0 is

$$(N2.47) \quad \Psi(x_0) = e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then the first component at x_0 is zero. After one period,

$$(N2.48) \quad \Psi(x_0 + L) = T(\lambda) \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Using (N2.46), this is

$$(N2.49) \quad T(\lambda)e_2 = \begin{pmatrix} T_{12}(\lambda) \\ T_{22}(\lambda) \end{pmatrix}.$$

The endpoint Dirichlet condition is therefore

$$(N2.50) \quad T_{12}(\lambda) = 0.$$

When this holds, e_2 is also an eigenvector of T , with multiplier $\kappa = T_{22}(\lambda)$. Hence an auxiliary spectral point is not only a value of λ ; one must also specify which sheet of the spectral curve is chosen, or equivalently which multiplier is attached to the zero. This is why the auxiliary spectrum is naturally a divisor on the spectral curve.

Bridge to the finite-gap construction. The periodic spectral theory gives two kinds of data. The main spectrum determines the spectral curve through the Floquet discriminant and the branch or double points. The auxiliary divisor determines the phase position on that curve. In finite-gap theory, the x - and t -dependence of the divisor becomes linear after the Abel transform, and the potential is reconstructed by theta functions. Those theta function formulas are not derived here; the goal of this note is only to set up the spectral objects used later in Section 3.

HW 4: Derivation of Formulas (18) and (19)

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